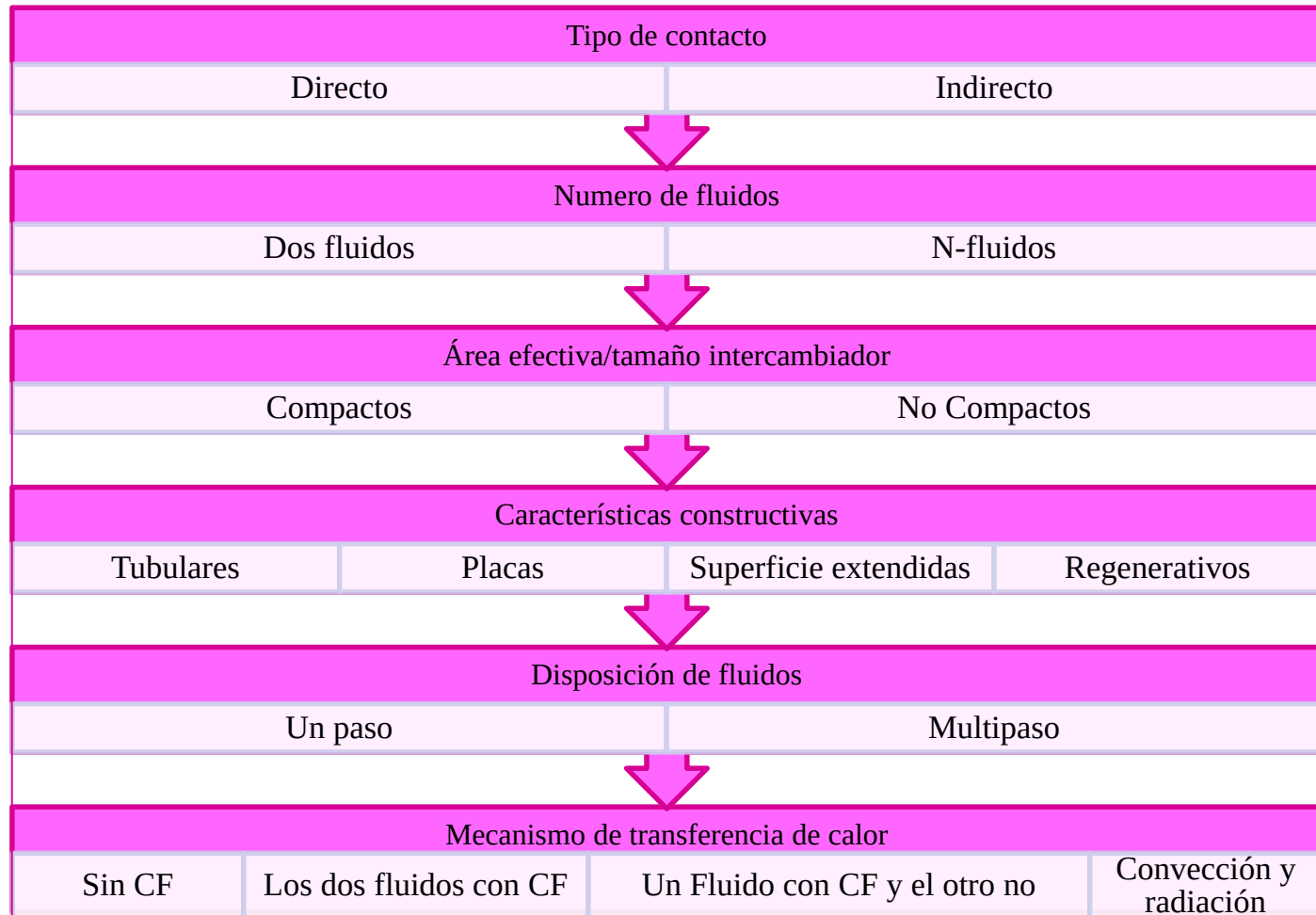


- Clasificación de intercambiadores
- Cálculo de intercambiadores
- Coste y selección de intercambiadores
- Configuraciones en serie de tubo-carcasa

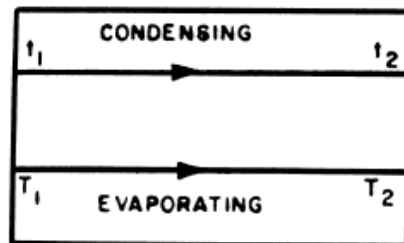
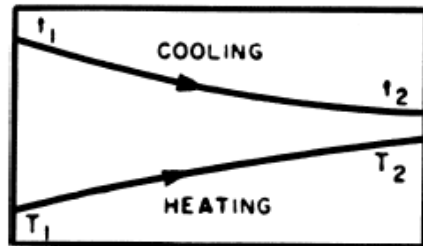
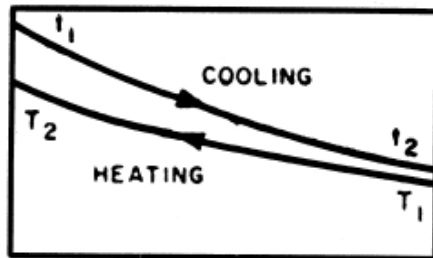
Sólo existen dos problemas a resolver con los intercambiadores y son los relativos a el diseño y la selección de los intercambiadores y a la operación de estos.

Para la selección dependerá de las características operativas que debe cumplir: presiones de trabajo, temperaturas, características de compatibilidad, etc... Una vez seleccionado su cálculo dependerá de los mecanismos de transferencia y de la disposición de los fluidos

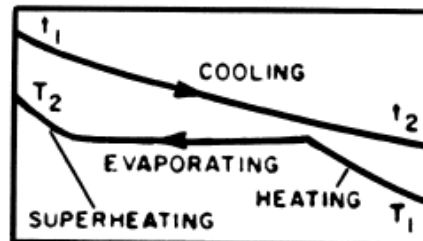
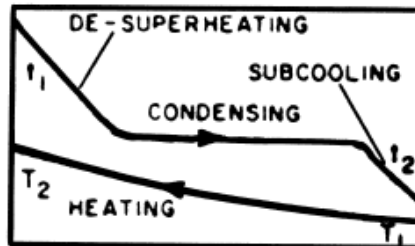
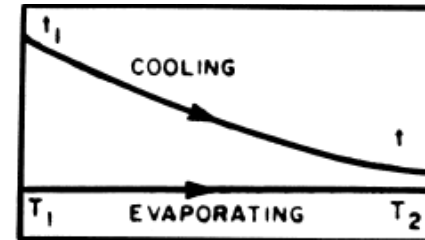
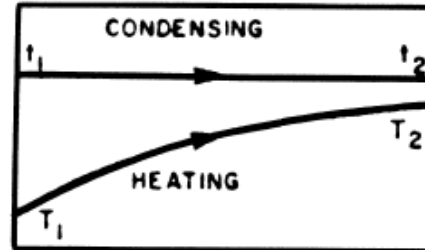


Mecanismos

Sin CF



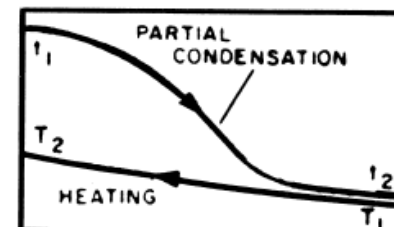
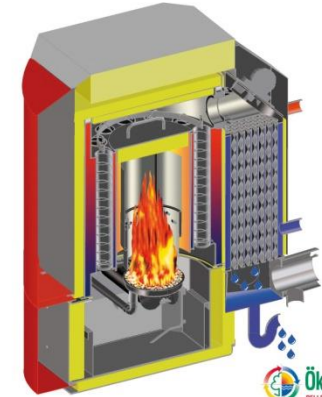
Los dos fluidos con CF



Un Fluido con CF y el otro no



Convección y radiación



Un paso

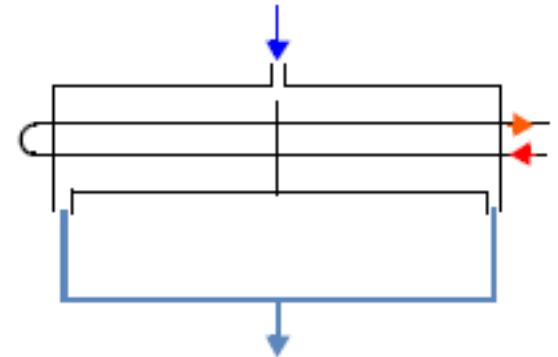
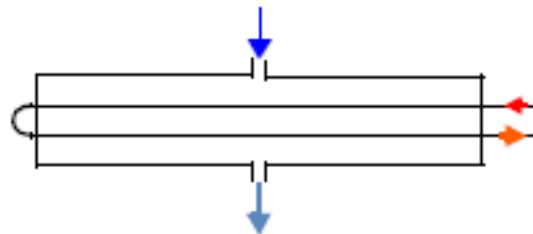
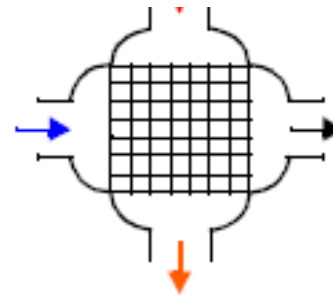
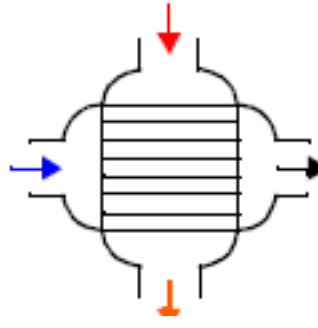
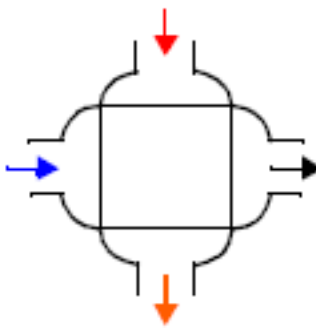
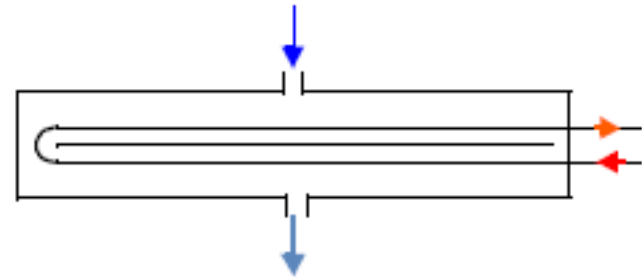
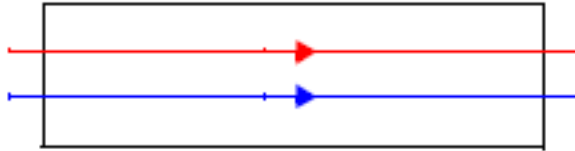
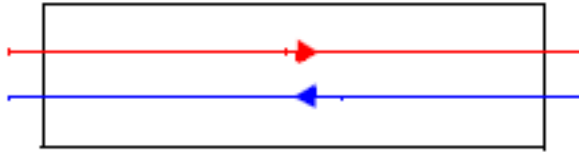
Contracorriente

Paralelo

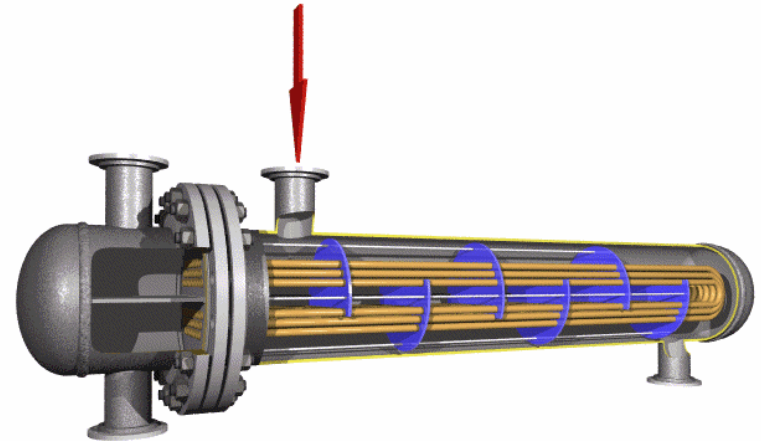
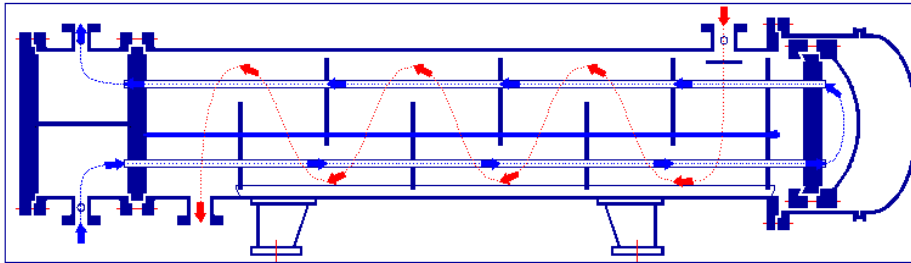
Cruzado

Partido

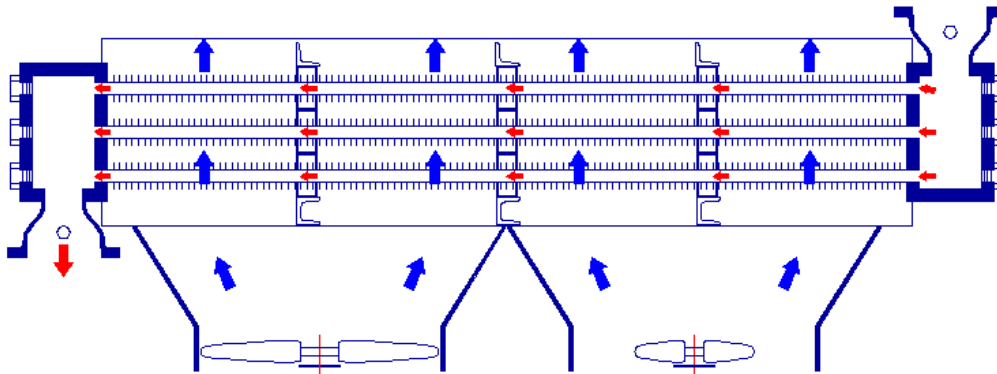
Dividido



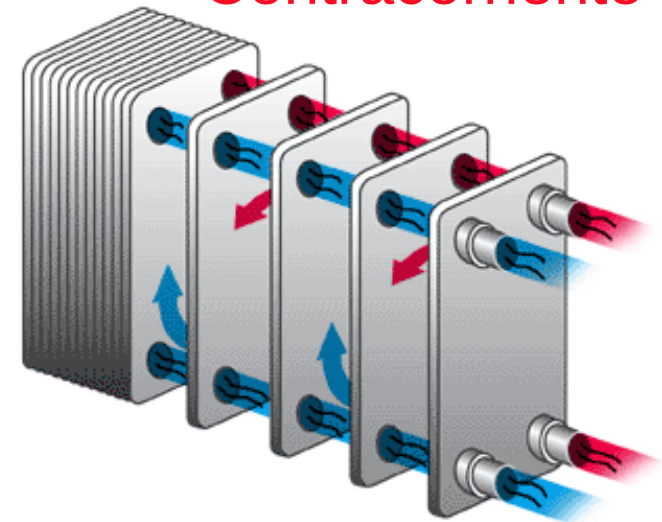
Multipaso (2 por los tubos)



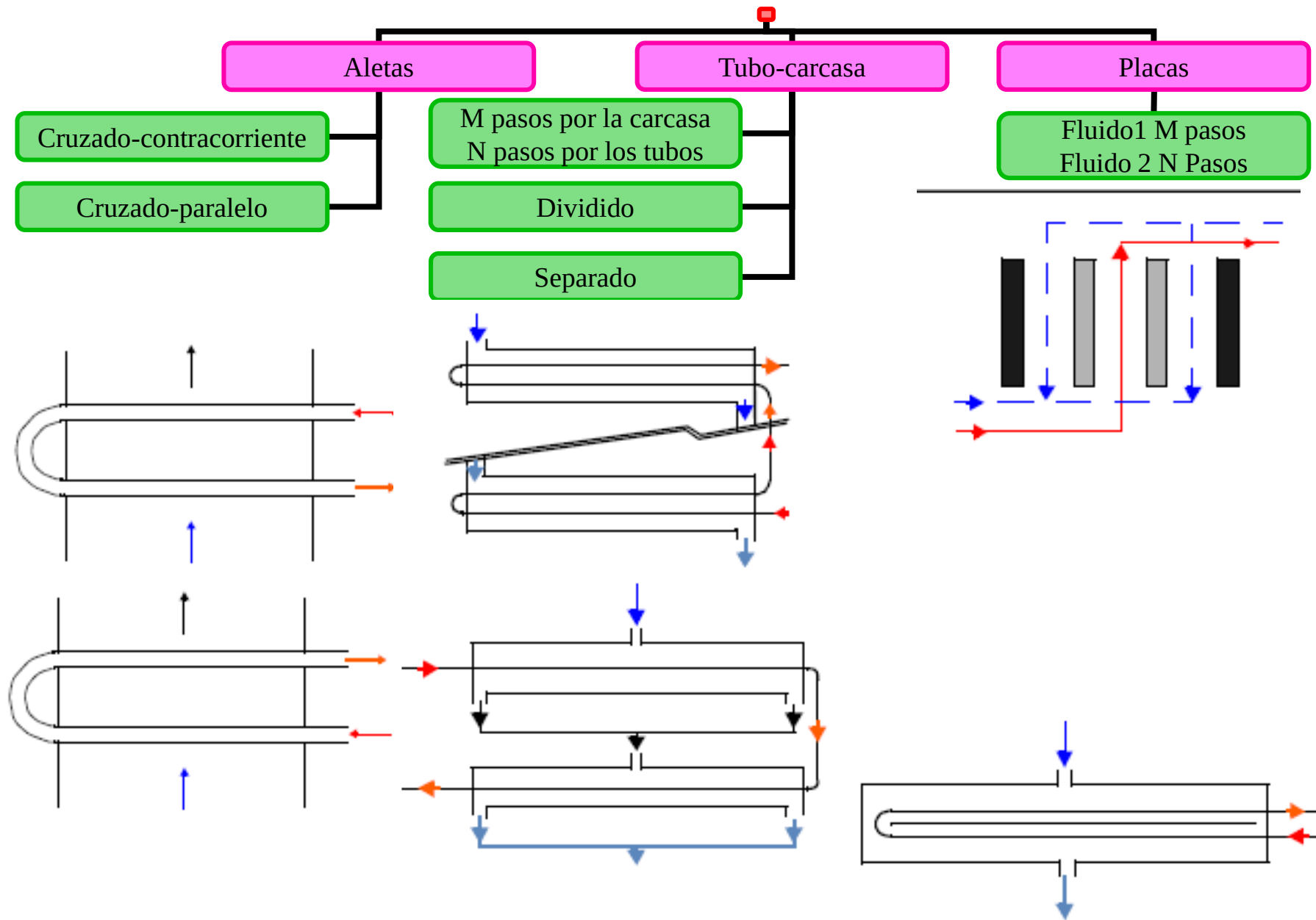
Cruzado

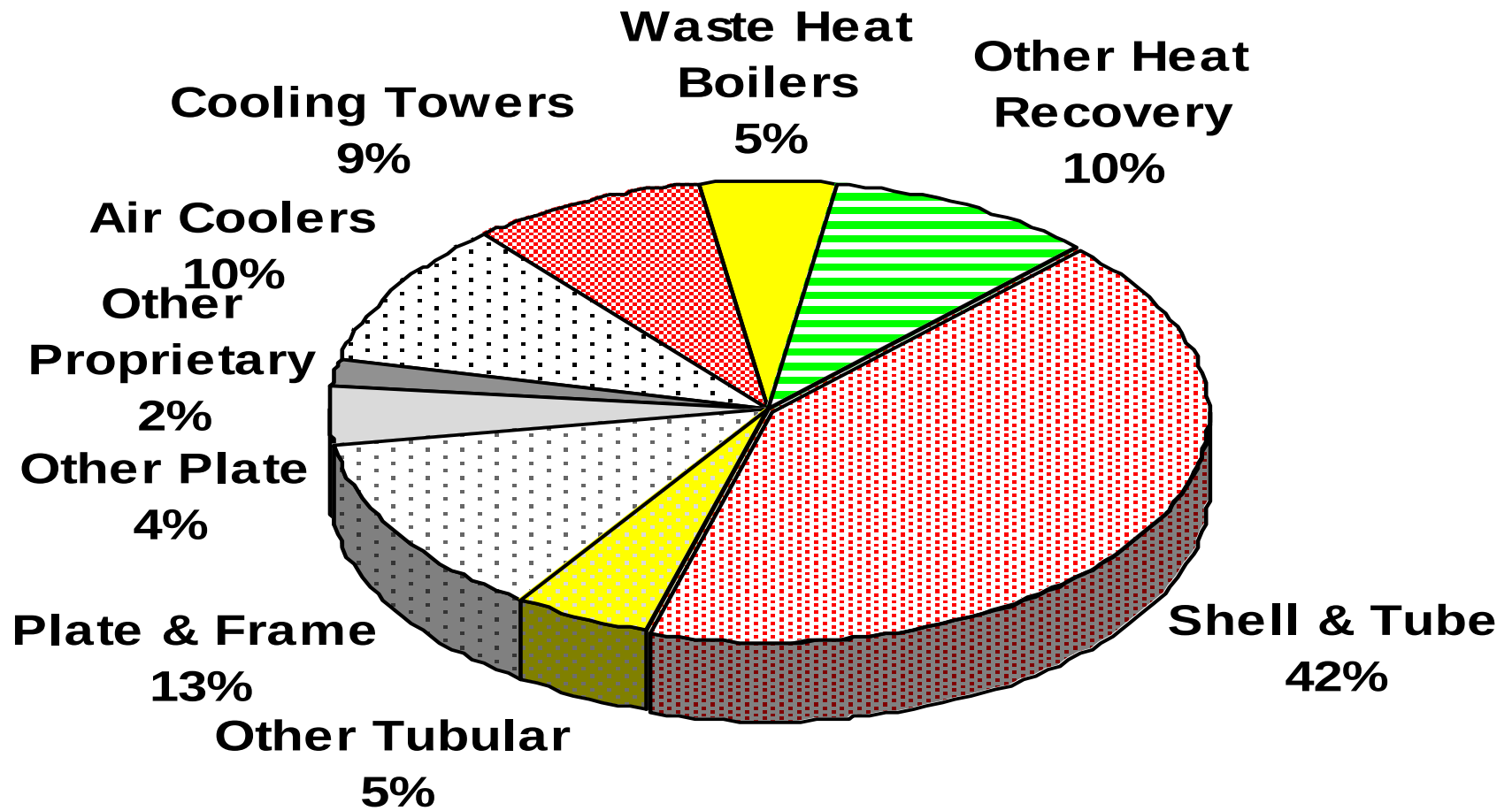


Contracorriente



Disposición de los Fluidos (multipaso)

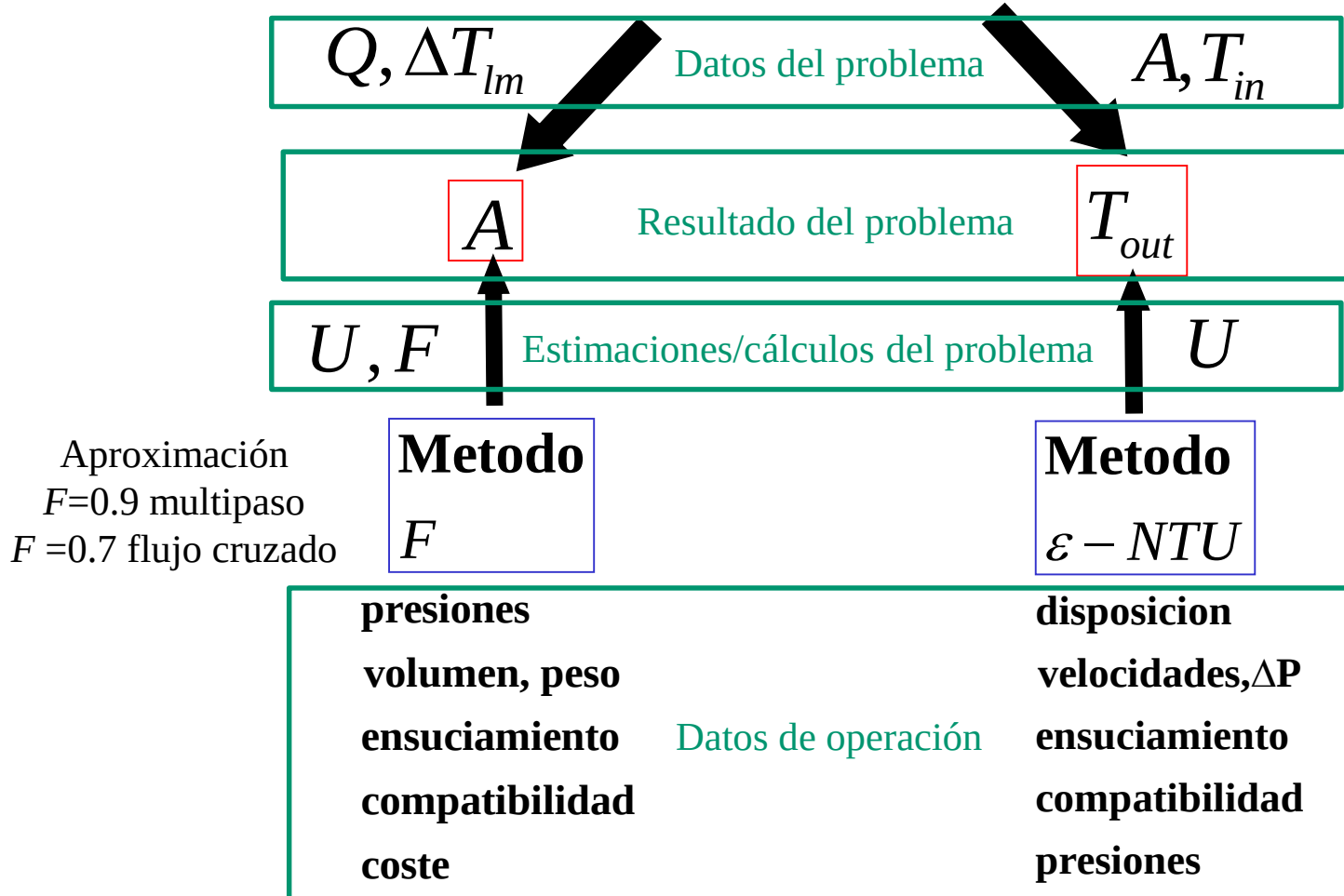


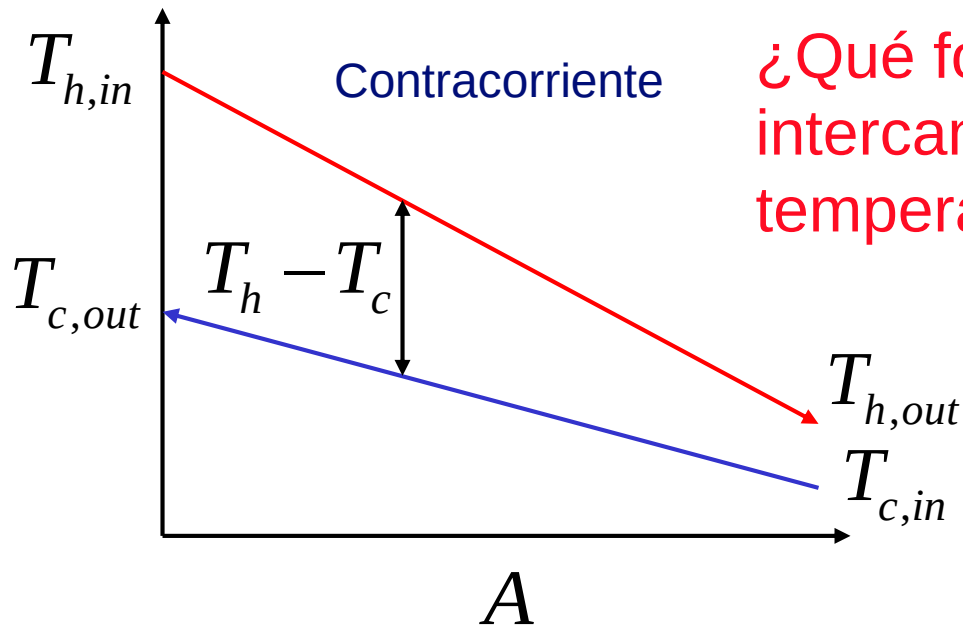


$$Q \Rightarrow mc_p \Delta T_H = mc_p \Delta T_C = mh_{fg}$$

$$F \Delta T_{lm}$$

$$Q = UA \Delta T_m$$





¿Qué fórmula me da el calor total intercambiado en función de las temperaturas de entrada y salida?

$$\frac{dQ}{dA} = -(mc_p)_h \frac{dT_h}{dA}$$

$$\frac{dQ}{dA} = -(mc_p)_c \frac{dT_c}{dA}$$

$$\frac{dQ}{dA} = U(T_h - T_c)$$

$U(x)$

$h_1(x)$

$h_2(x)$

$\frac{dQ}{dA} = U(x)(T_h(x) - T_c(x))$

$$\frac{dQ}{dA} \left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) = \frac{d(T_h - T_c)}{dA}$$

Si el coeficiente no depende de la posición lo que ocurre globalmente también ocurre localmente

$$\left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) dA = \frac{d(T_h - T_c)}{U(T_h - T_c)}$$

$$\int_0^A \left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) dA = \int_{T_{h,in}-T_{c,out}}^{T_{h,out}-T_{c,in}} \frac{d(T_h - T_c)}{U(T_h - T_c)} \quad \left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) A = \frac{1}{U} \ln \left(\frac{T_{h,out} - T_{c,in}}{T_{h,in} - T_{c,out}} \right)$$

$$Q = (mc_p)_c \Delta T_c = (mc_p)_h \Delta T_h \Rightarrow Q \left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) = \Delta T_c - \Delta T_h$$

Para un intercambiador en Contracorriente: La media de diferencias de temperaturas es el incremento de temperaturas logarítmico medio

$$Q = UA \frac{\overbrace{\Delta T_h - \Delta T_c}^{\Delta T_m}}{\ln \left(\frac{T_{h,in} - T_{c,out}}{T_{h,out} - T_{c,in}} \right)} = UA \Delta T_{lm}$$

Para cualquier otro intercambiador con diferente disposición de los fluidos su diferencia media de temperaturas se compara con el de contracorriente

contracorriente

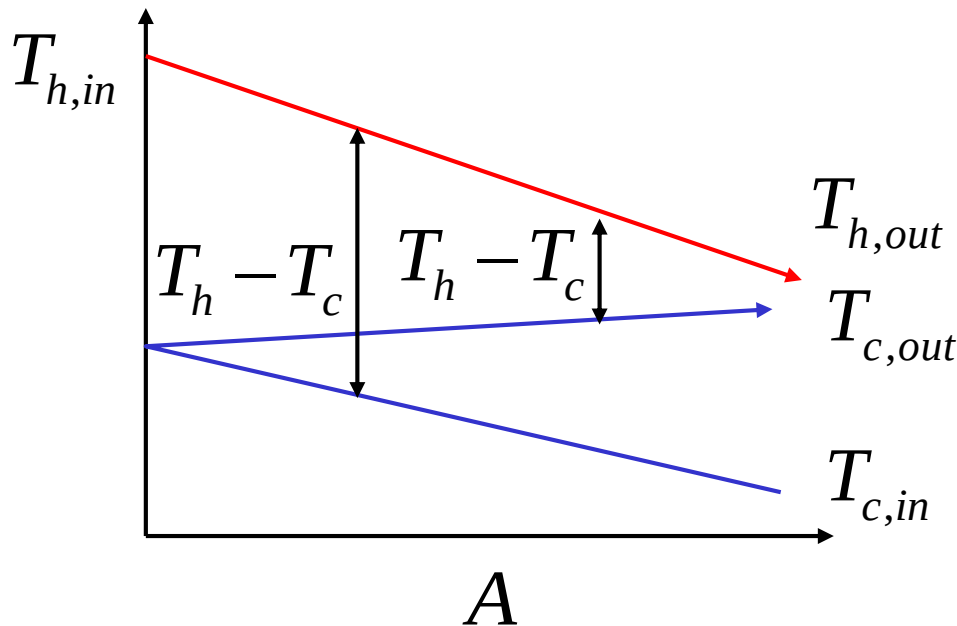
$$Q = UA\Delta T_m = UA\Delta T_{lm}$$

Otra disposición

$$Q = UA\Delta T_{m,otra}$$

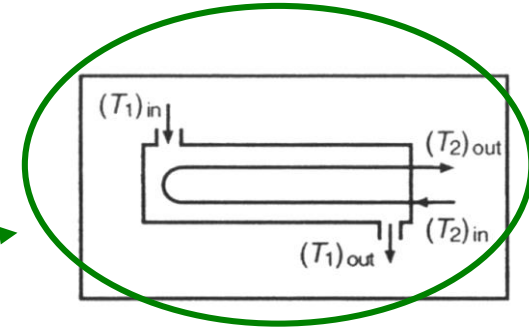
$$Q = UA \underbrace{\frac{\Delta T_{m,otra}}{\Delta T_{lm}}}_F \Delta T_{lm}$$

$$F \leq 1$$



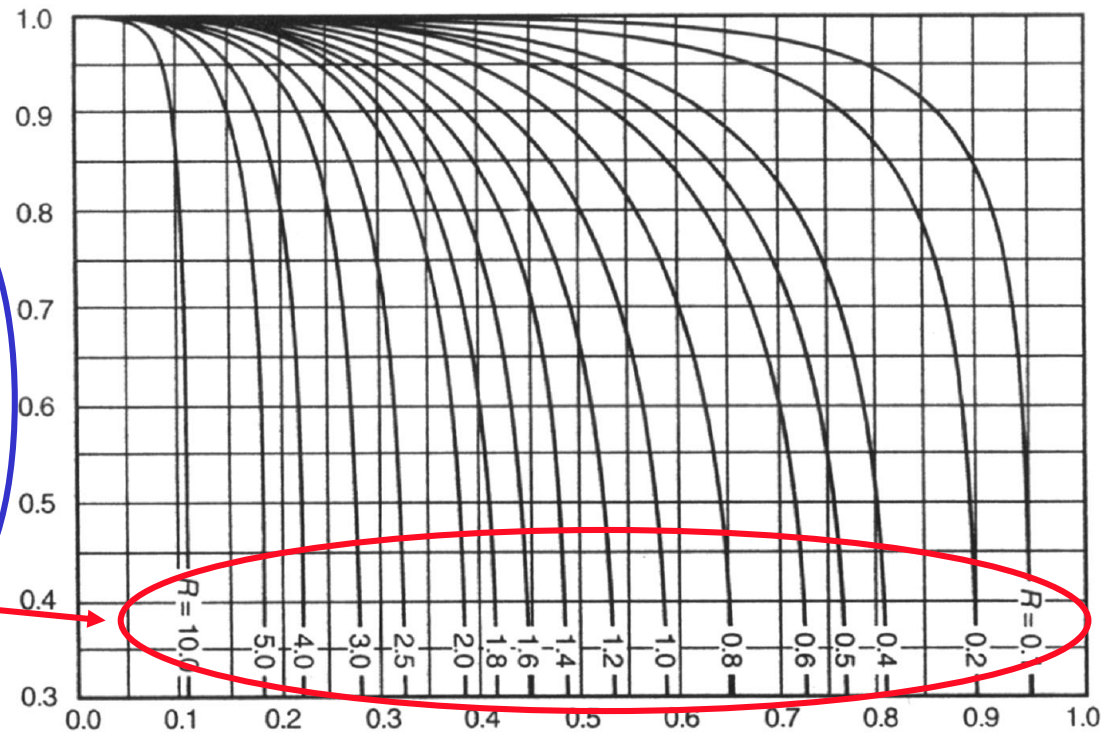
$$P = \frac{\Delta T_2}{\Delta T_{\max}}$$

$$R = \frac{C_2}{C_1} = \frac{(mc_p)_2}{(mc_p)_1} = \frac{\Delta T_1}{\Delta T_2}$$



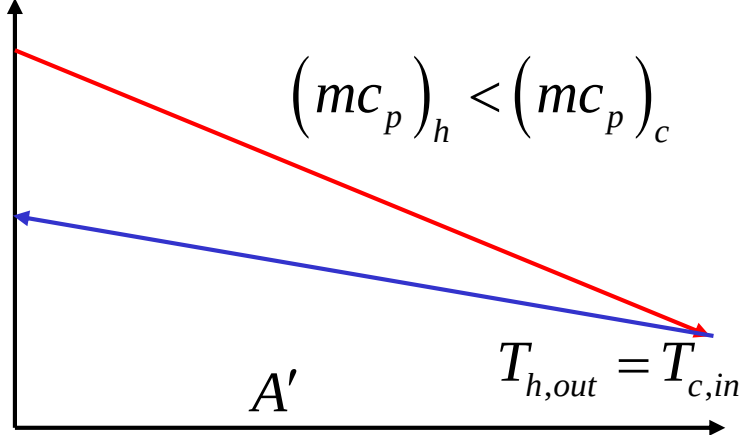
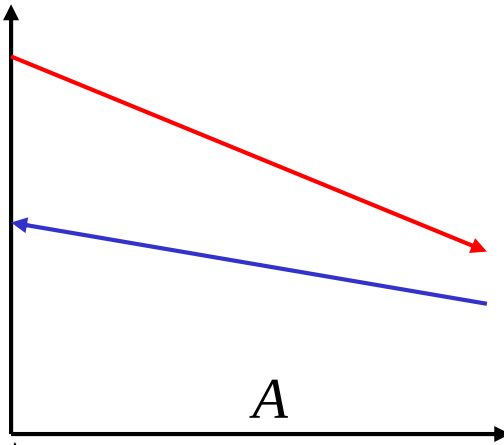
Para cada tipo de intercambiador se grafica F frente a la relación de capacidades (R) y a la eficiencia térmica (P)

$$F \text{ Correction Factor} = \frac{\Delta T_m}{\Delta T_{lm}}$$



$$P \text{ Thermal Effectiveness} = \frac{(T_2)_{\text{out}} - (T_2)_{\text{in}}}{(T_1)_{\text{in}} - (T_2)_{\text{in}}}$$

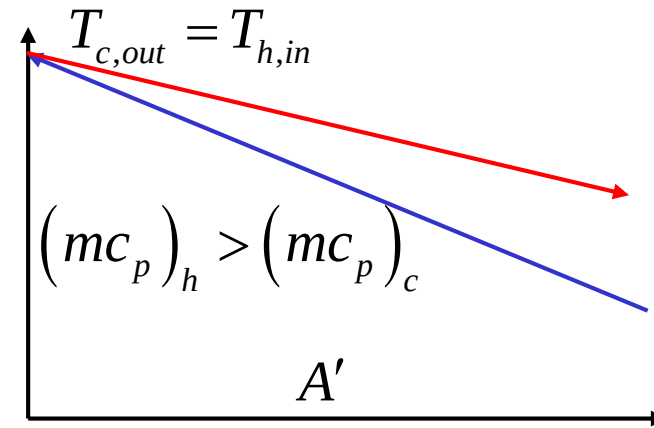
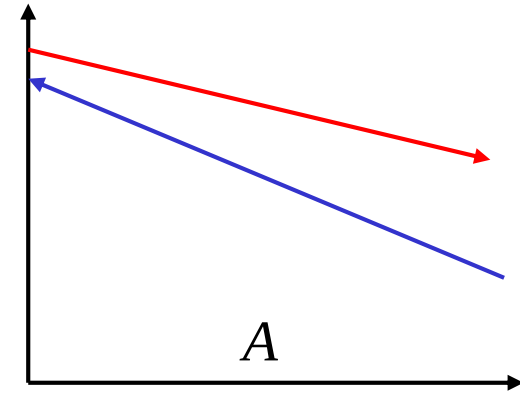
$$\varepsilon = \frac{Q}{Q_{\max}}$$



$$Q_{\max} = (mc_p)_h (T_{h,in} - T_{h,out}) = (mc_p)_h (T_{h,in} - T_{c,in})$$

$$Q_{\max} = (mc_p)_{\min} (T_{h,in} - T_{c,in})$$

$$Q_{\max} = (mc_p)_c (T_{c,out} - T_{c,in}) = (mc_p)_c (T_{h,in} - T_{c,in})$$



$$\varepsilon = P \text{ si } (mc_p)_2 = (mc_p)_{\min} ; \varepsilon = PR \text{ si } (mc_p)_1 = (mc_p)_{\min}$$

$$\left(\frac{1}{(mc_p)_2} - \frac{1}{(mc_p)_1} \right) AU = \ln \left(\frac{T_{1,out} - T_{2,in}}{T_{1,in} - T_{2,out}} \right)$$

$$\left(1 - \frac{\overbrace{(mc_p)_2}^R}{(mc_p)_1} \right) \frac{\overbrace{AU}^{NTU_2}}{(mc_p)_2} = \ln \left(\frac{T_{1,out} - T_{2,in}}{T_{1,in} - T_{2,out}} \right)$$

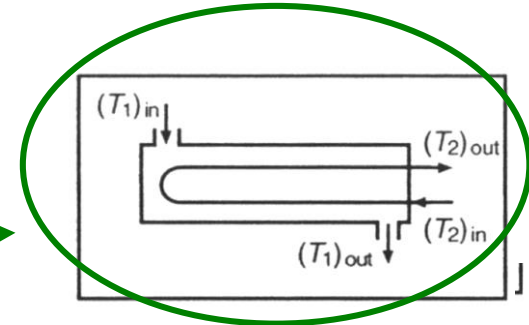
$$\frac{T_{2,out} - T_{2,in}}{T_{1,in} - T_{2,in}} = P; \frac{T_{1,in} - T_{2,out}}{T_{1,in} - T_{2,in}} = 1 - P; \frac{T_{1,out} - T_{2,in}}{T_{1,in} - T_{2,in}} = 1 - RP$$

$$\exp \left[(1 - R) NTU \right] = \frac{1 - RP}{1 - P}$$

$$f(R, NTU, P) = 0 \text{ otras configuraciones}$$

$$\theta = \frac{F \Delta T_{lm}}{\Delta T_{max}}$$

$$NTU_2 = \frac{UA}{C_2}$$



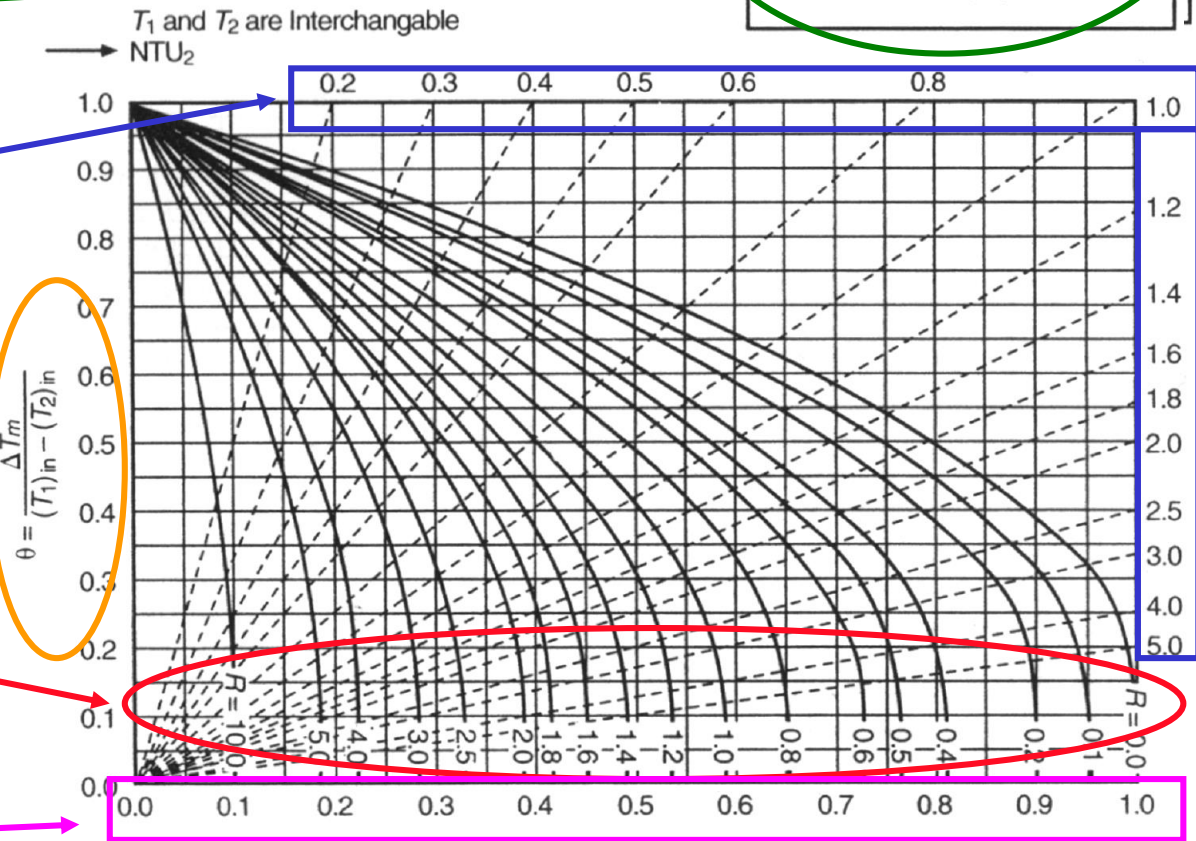
Para cada tipo de
intercambiador
se grafica
 NTU

y θ

frente a la
relación de
capacidades

(R) y
a la eficiencia
térmica

(P)



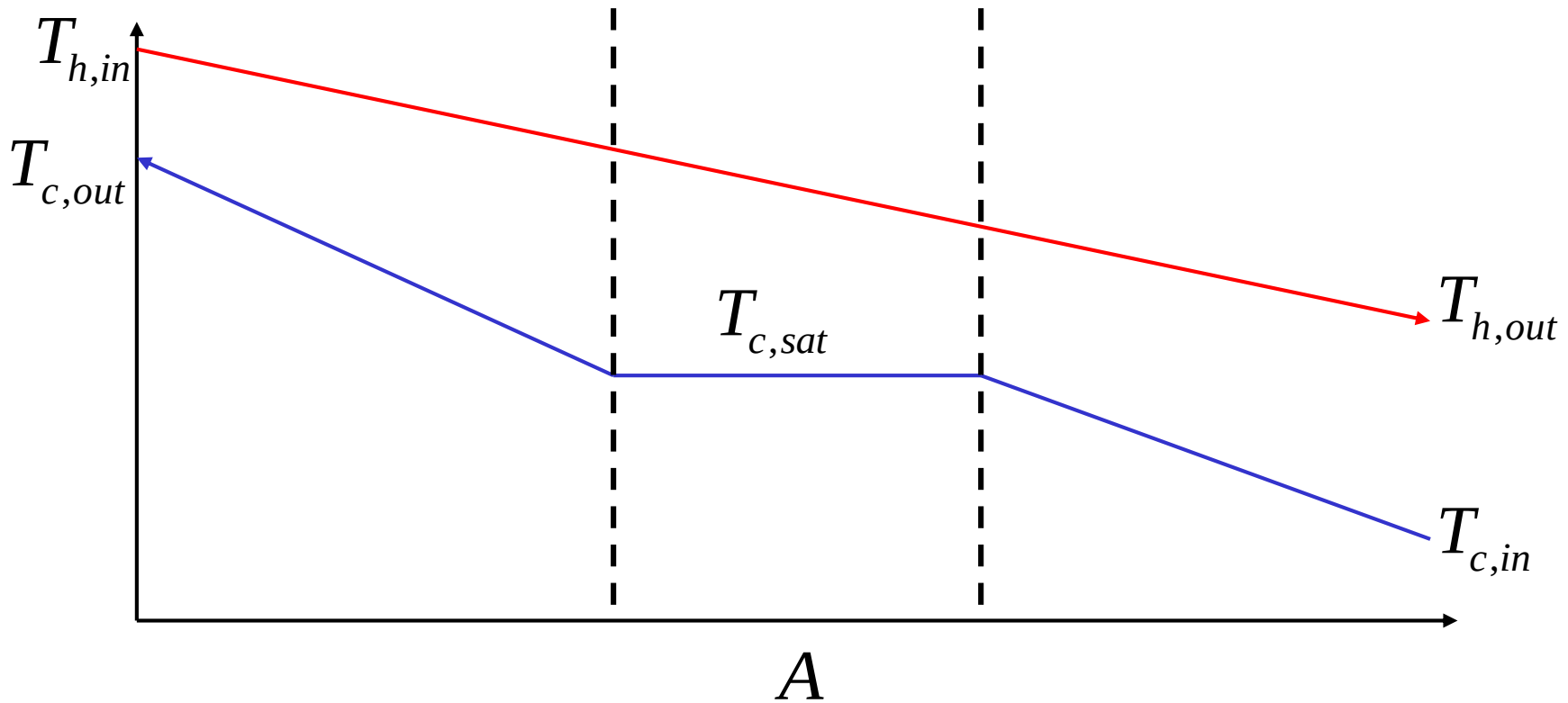
Hipótesis de los métodos de cálculo

- Pérdidas con el entorno despreciables
- c_p constantes
- U constante a lo largo del intercambiador
- T^a uniforme en la dirección del flujo
- Cuando existen recorridos paralelos los gastos y las áreas de transferencia son iguales en todos los caminos

Validez y aproximaciones de los métodos de cálculo

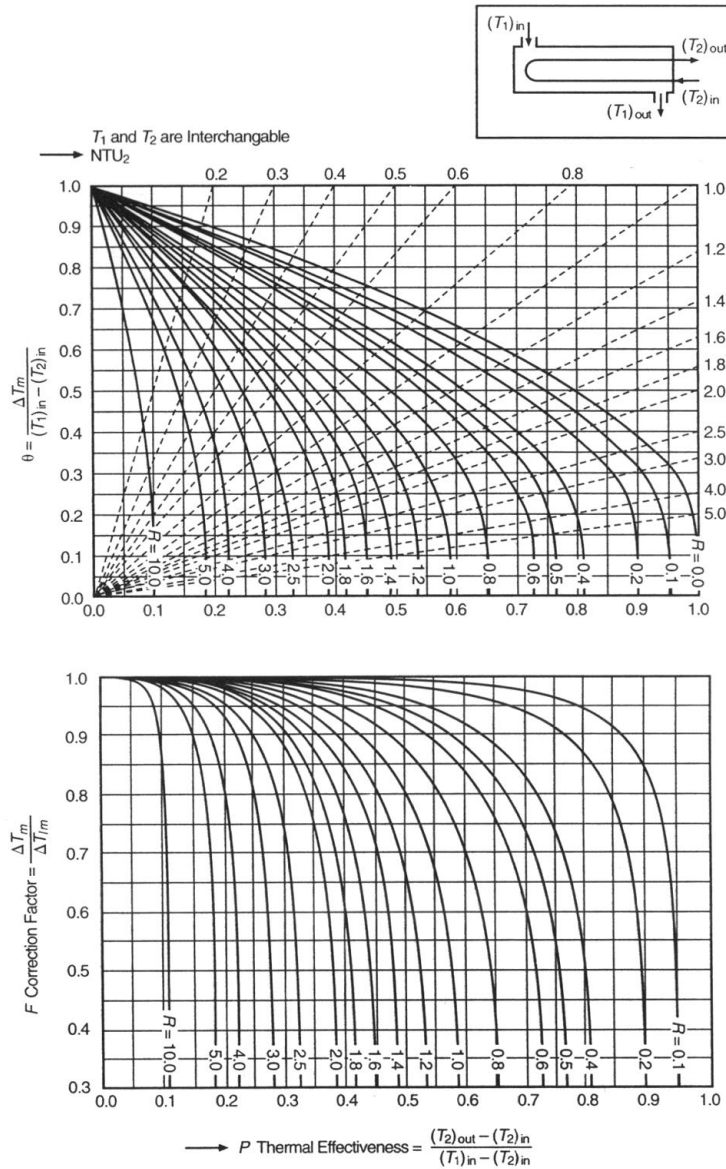
Fluido frío	Fluido caliente	Nota
1 fase	1 fase	Variaciones de c_p con la T^a pequeñas
evaporándose	1 fase	$F=1$ $C=0$ (independientemente del intercambiador)
1 fase	condensándose	$F=1$ $C=0$ (independientemente del intercambiador)
Evaporándose una mezcla	1 fase	Si el cambio de fase no es significativo, U es cte y c_p es una función del calor latente y sensible
1 fase	Condensándose una mezcla	Si el cambio de fase no es significativo, U es cte y c_p es una función del calor latente y sensible

Quando NO se cumplen las hipótesis de los modelos se trata de dividir el intercambiador en subintercambiadores que sí las cumplen

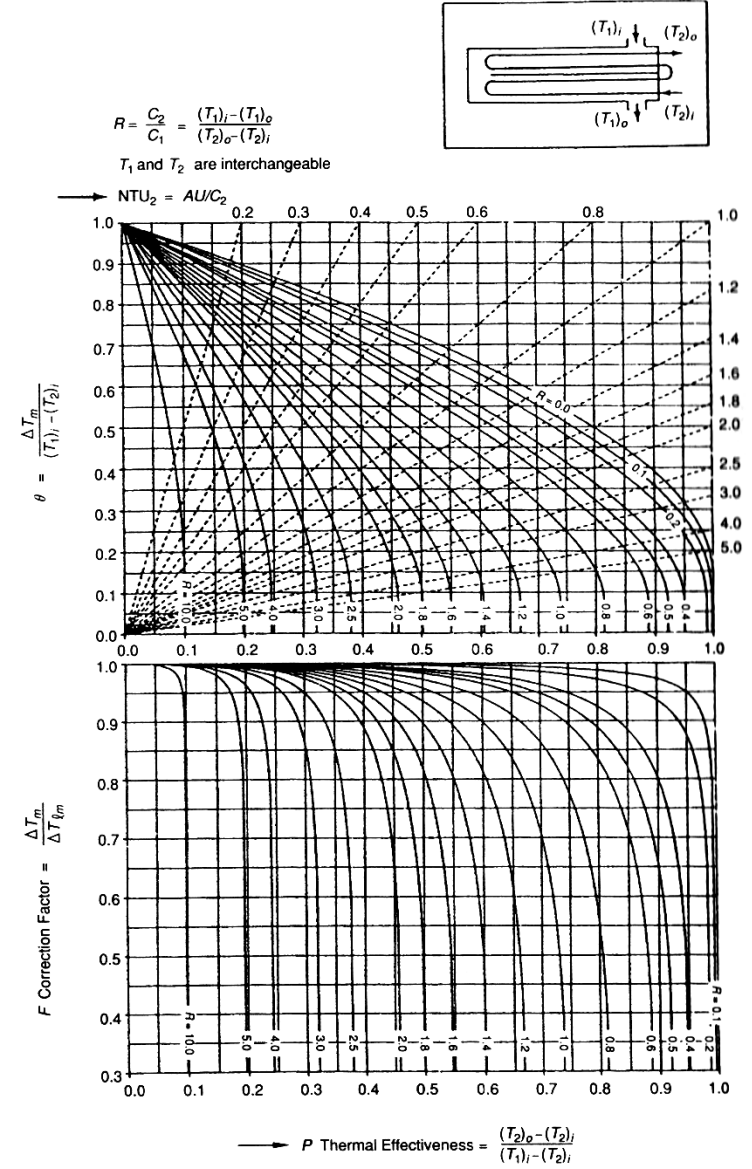


Otros casos

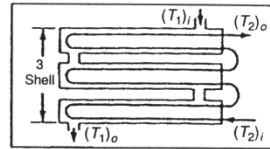
$$Q_i = A_i U_i \Delta T_{m,i} \quad Q = \sum Q_i$$



Tubo carcasa tipo **E** un paso por la carcasa y un nº par de pasos por los tubos

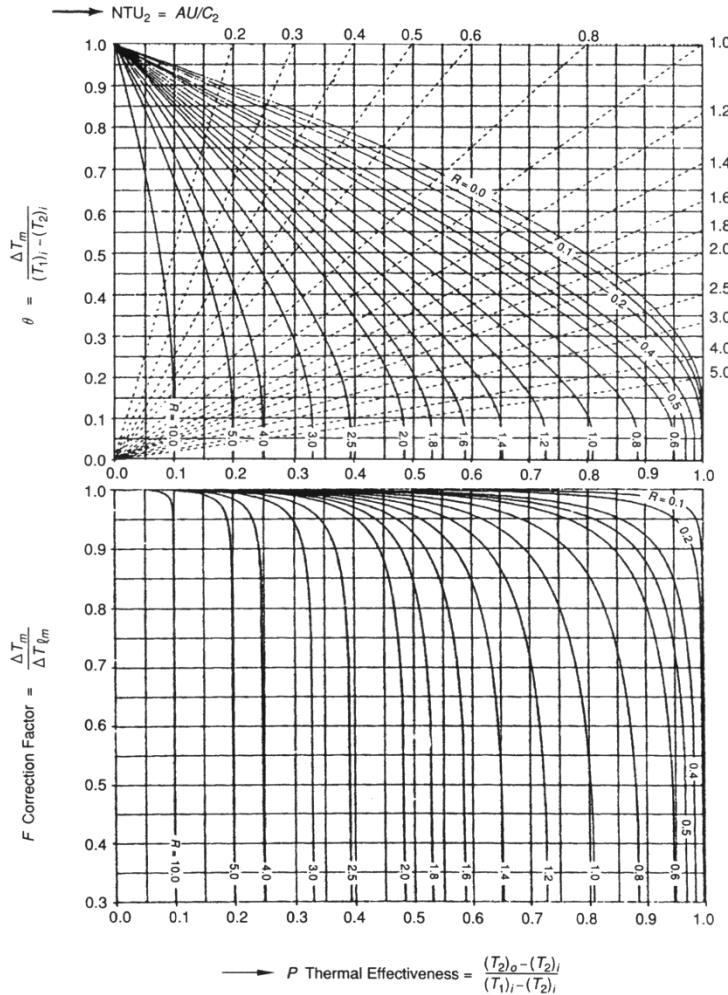


Tubo carcasa tipo **E** dos pasos por la carcasa y un nº par de pasos por los tubos

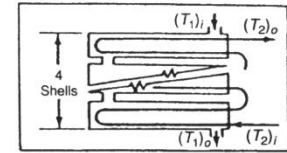


$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

T_1 and T_2 are interchangeable

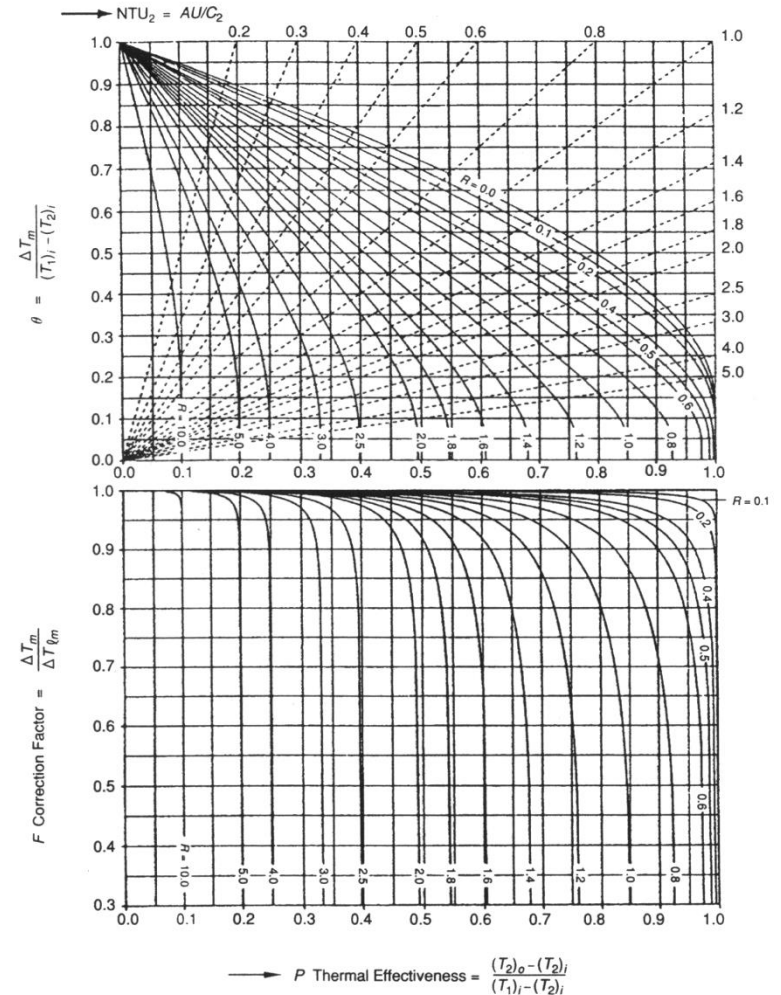


Tubo carcasa tipo **E** tres pasos por la carcasa y un nº par de pasos por los tubos

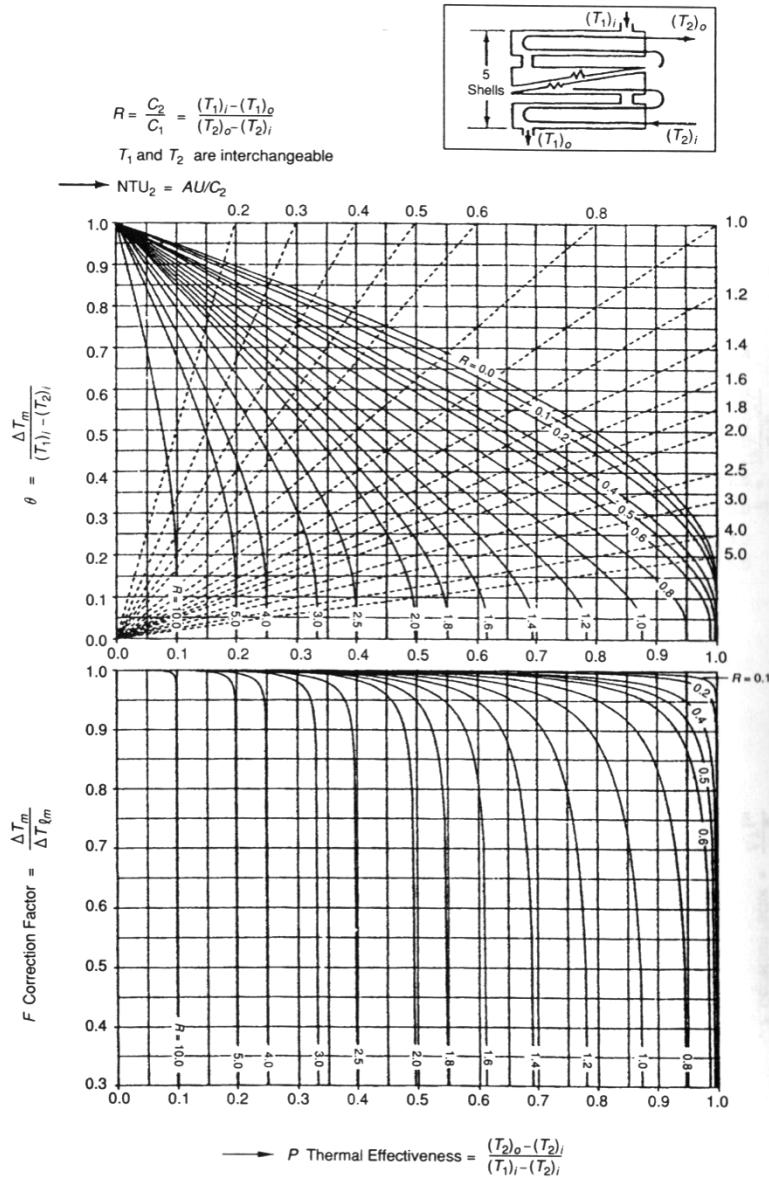


$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

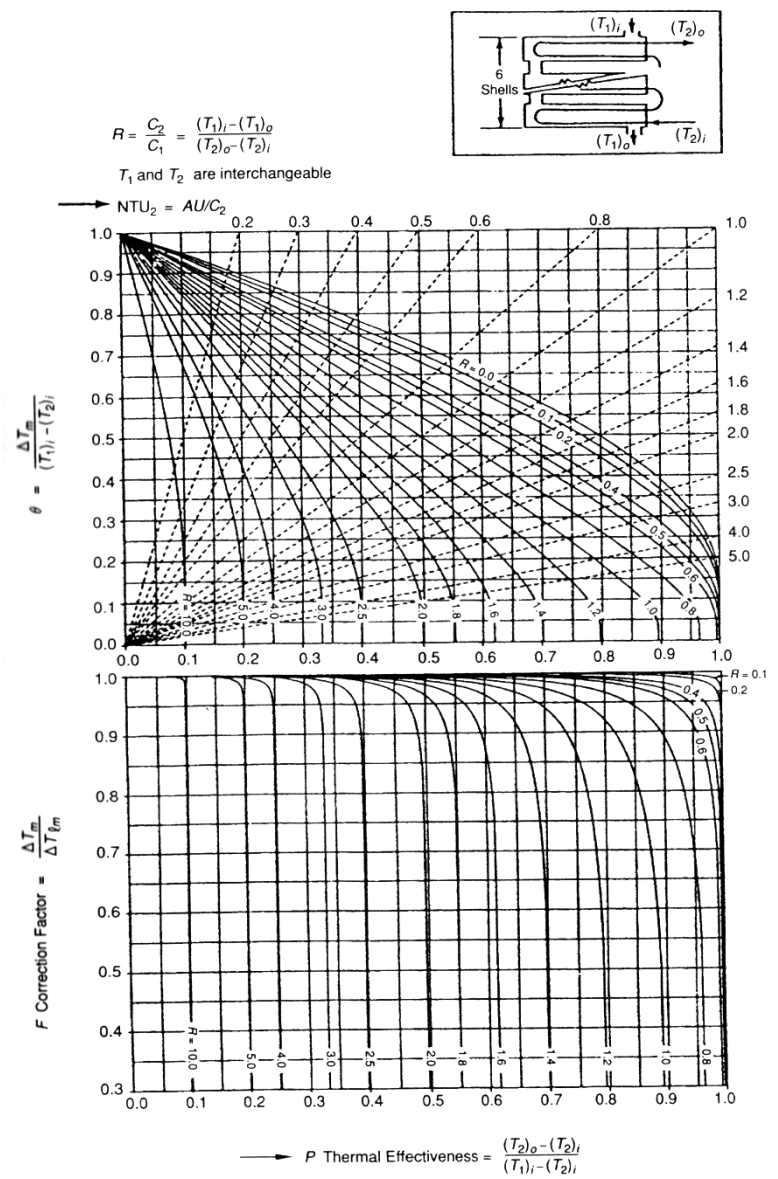
T_1 and T_2 are interchangeable



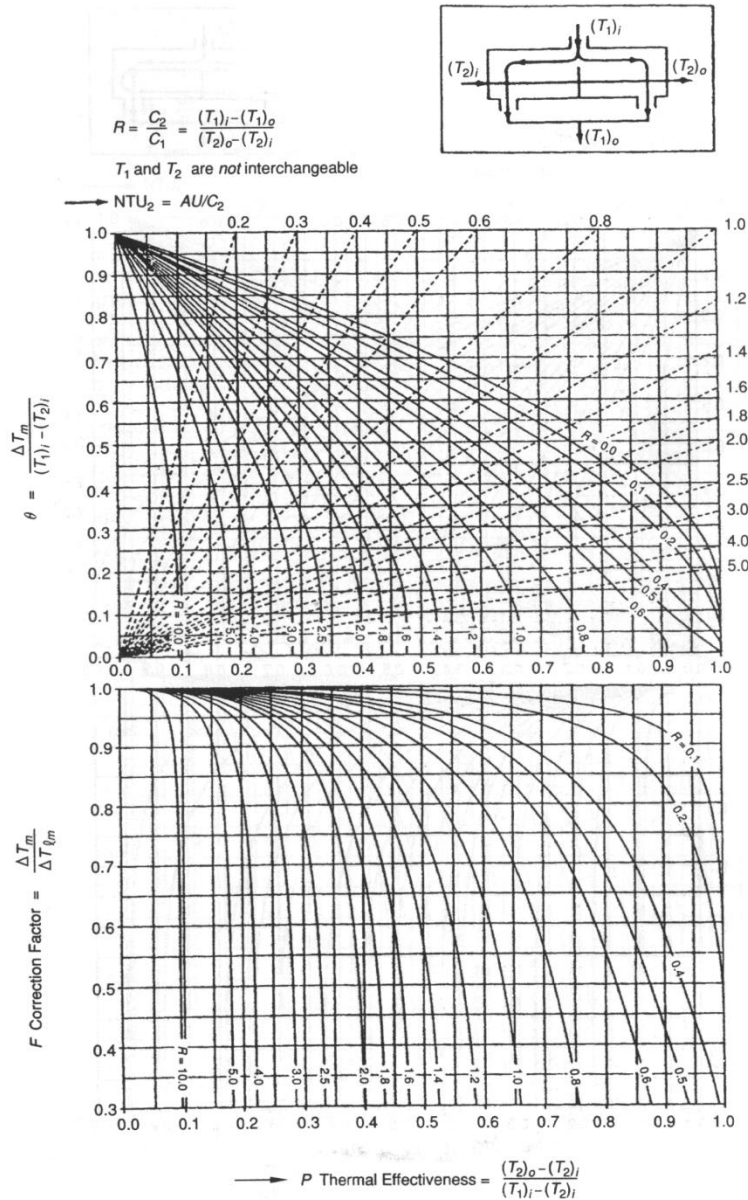
Tubo carcasa tipo **E** cuatro pasos por la carcasa y un nº par de pasos por los tubos



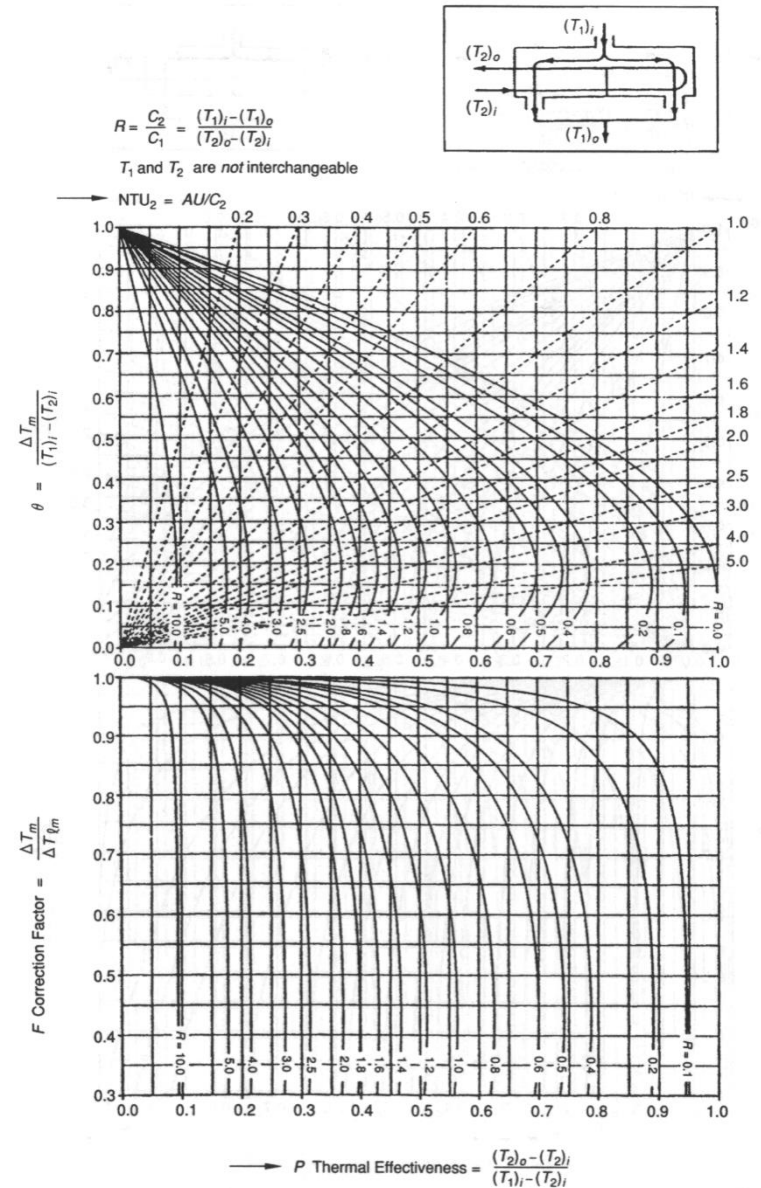
Tubo carcasa tipo **E cinco** pasos por la carcasa y un nº par de pasos por los tubos



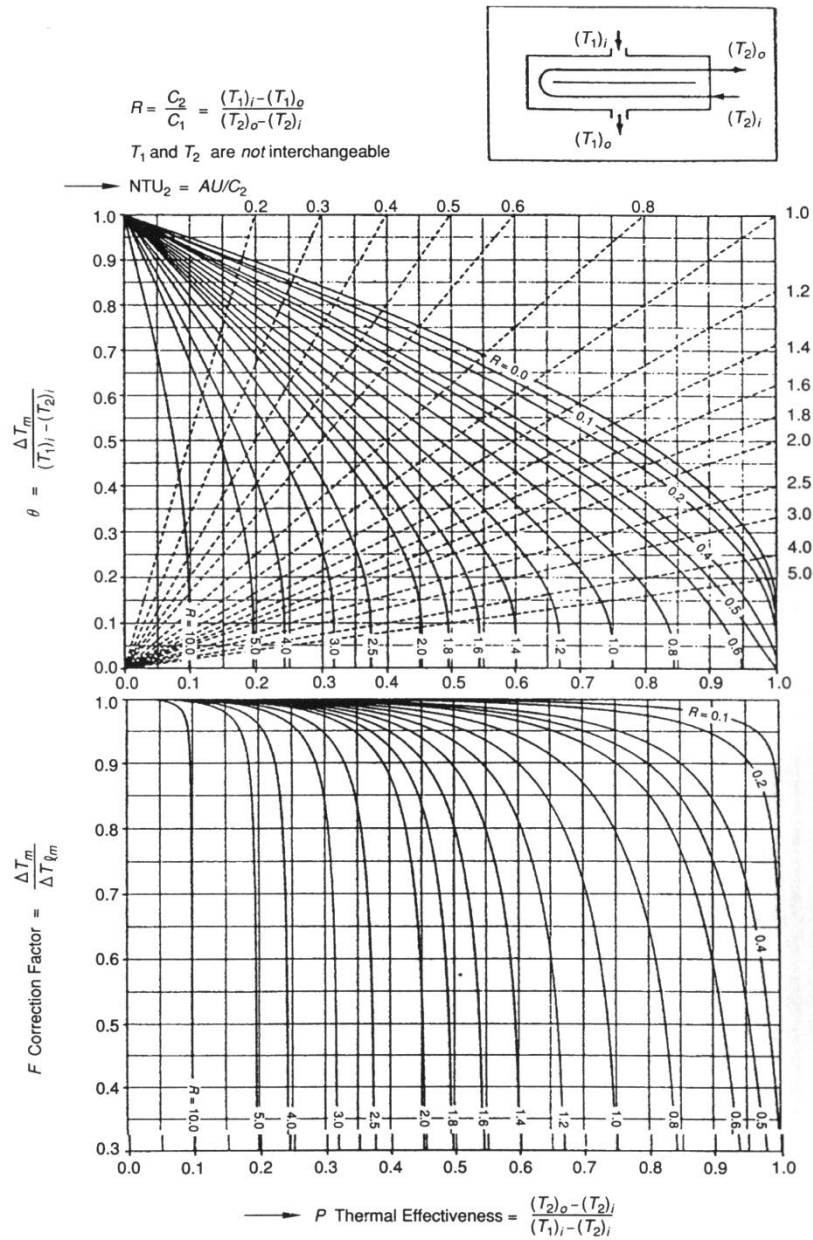
Tubo carcasa tipo **E seis** pasos por la carcasa y un nº par de pasos por los tubos



Tubo carcasa tipo **J** un paso por los tubos

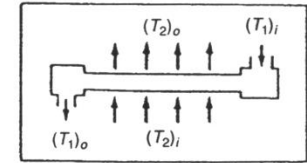
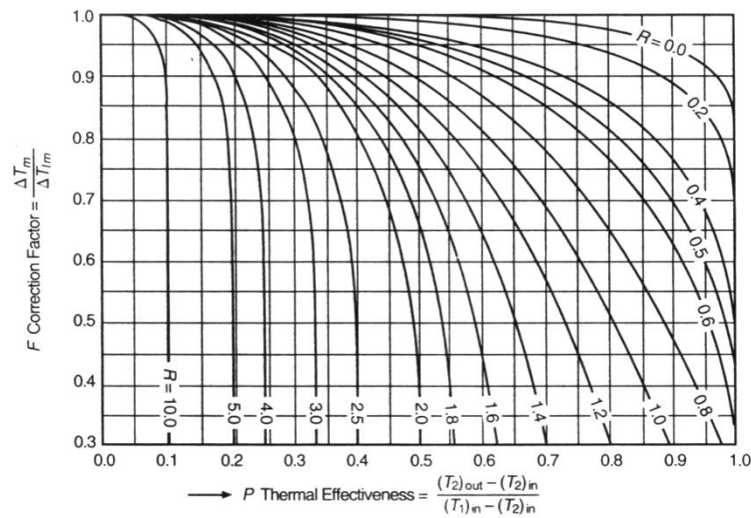
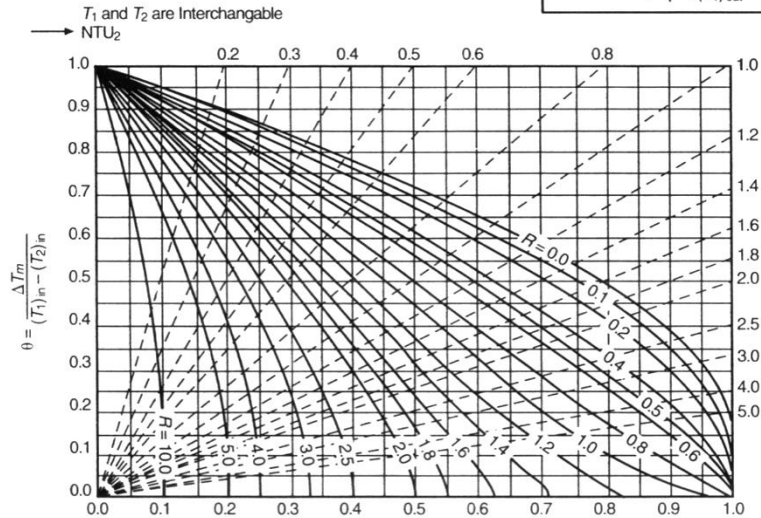
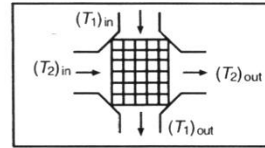


Tubo carcasa tipo **J** dos pasos por los tubos



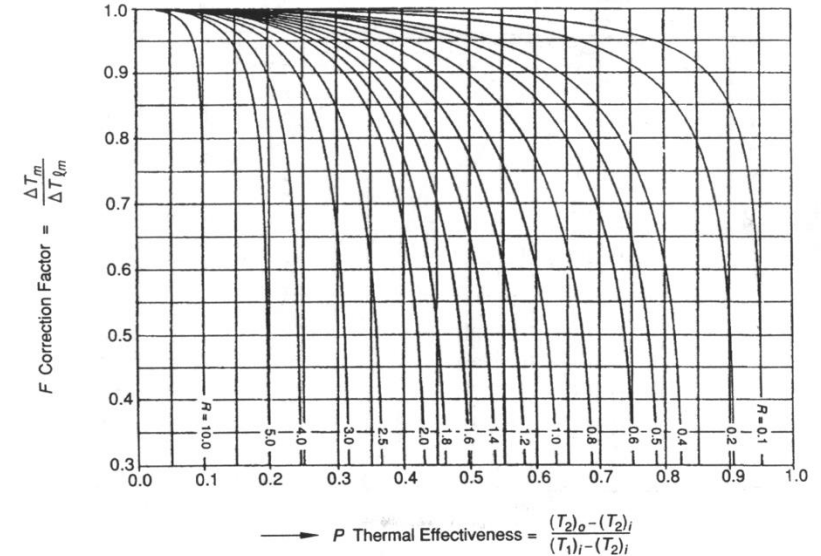
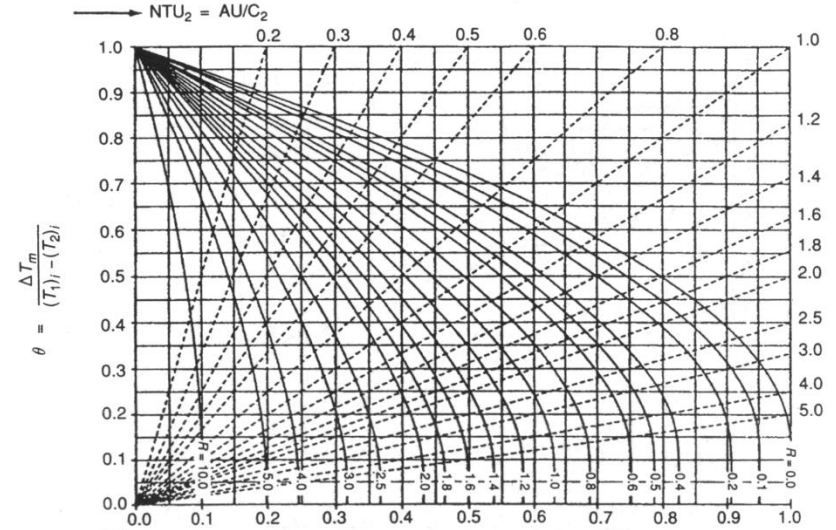
Tubo carcasa tipo **G** y un nº par de pasos por los tubos

Cruzado (no mezclados/1F 1P no mezclado)



$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

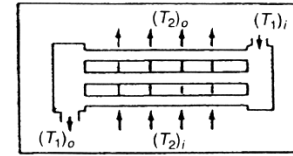
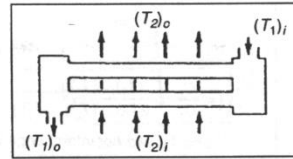
T_1 and T_2 are not interchangeable



Cruzado (2F 1P no mezclado /3F 1P no mezclado)

$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

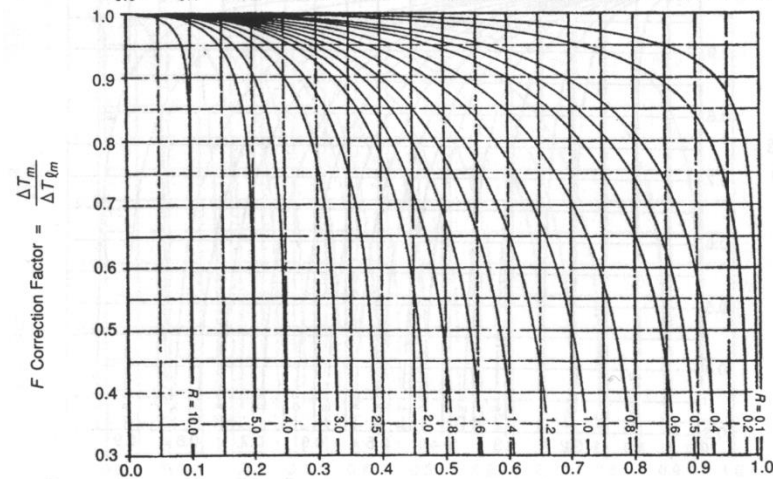
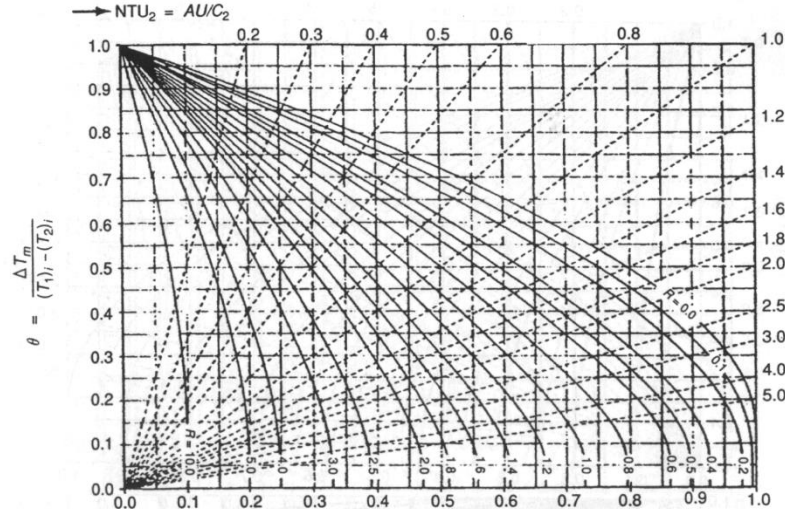
T_1 and T_2 are not interchangeable



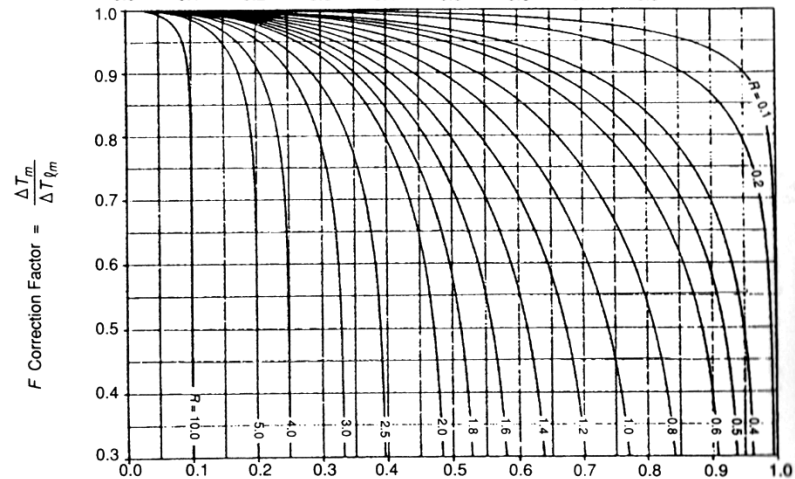
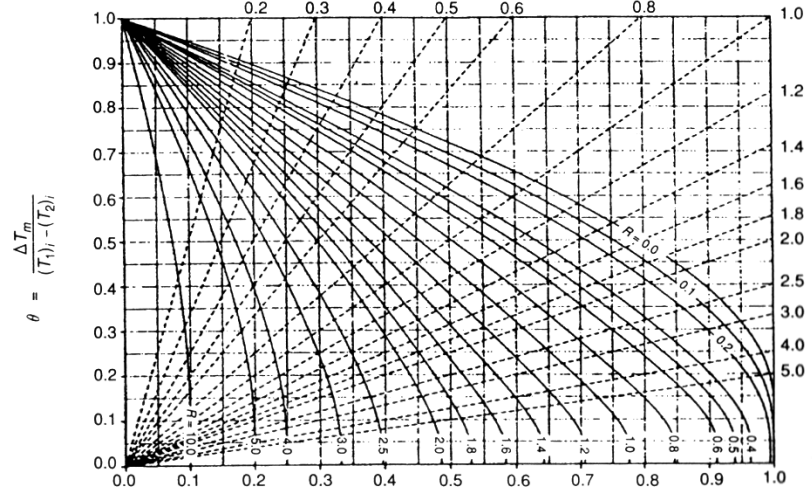
$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

T_1 and T_2 are not interchangeable

$$NTU_2 = AU/C_2$$



$$P \text{ Thermal Effectiveness} = \frac{(T_2)_o - (T_2)_i}{(T_1)_i - (T_2)_i}$$



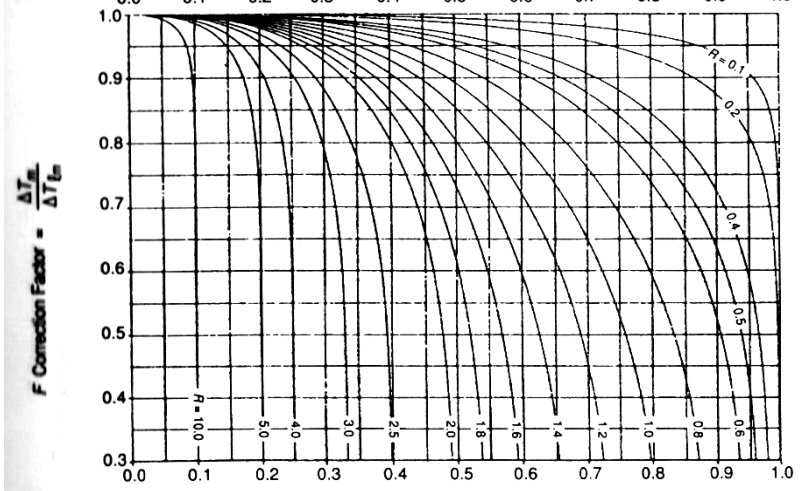
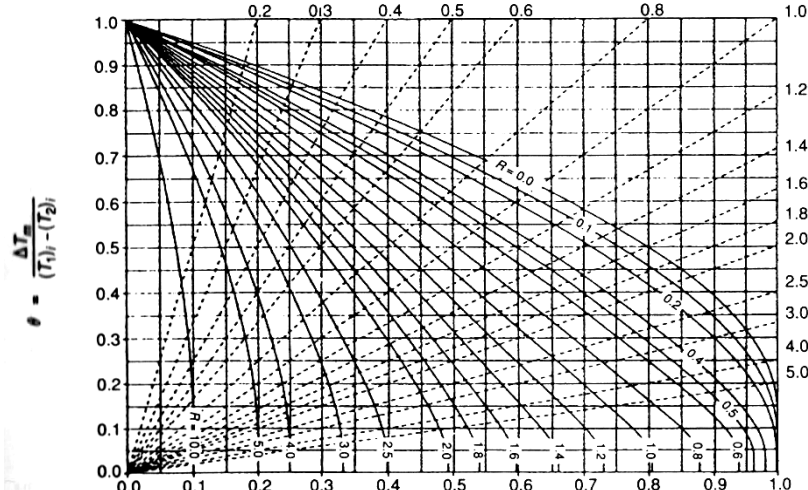
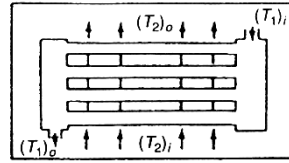
$$P \text{ Thermal Effectiveness} = \frac{(T_2)_o - (T_2)_i}{(T_1)_i - (T_2)_i}$$

Cruzado (4F 1P no mezclado / 2F 2P no mezclado entre pasos)

$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

T_1 and T_2 are not interchangeable

$$\rightarrow NTU_2 = AU/C_2$$

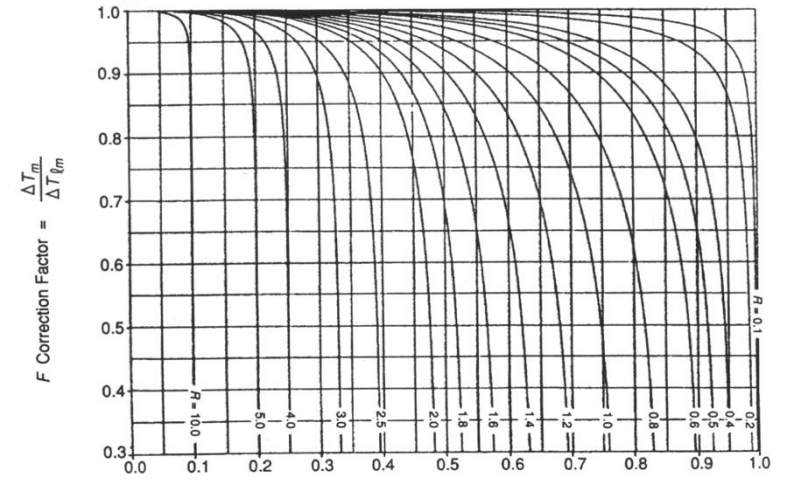
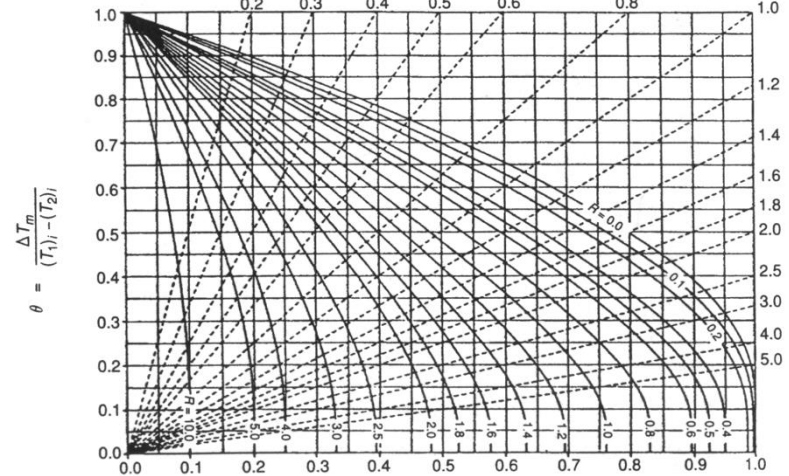
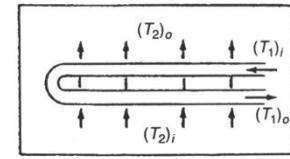


$$\rightarrow P \text{ Thermal Effectiveness} = \frac{(T_2)_o - (T_2)_i}{(T_1)_i - (T_2)_i}$$

$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

T_1 and T_2 are not interchangeable

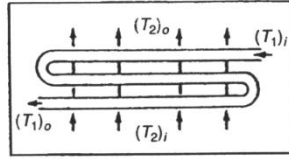
$$\rightarrow NTU_2 = AU/C_2$$



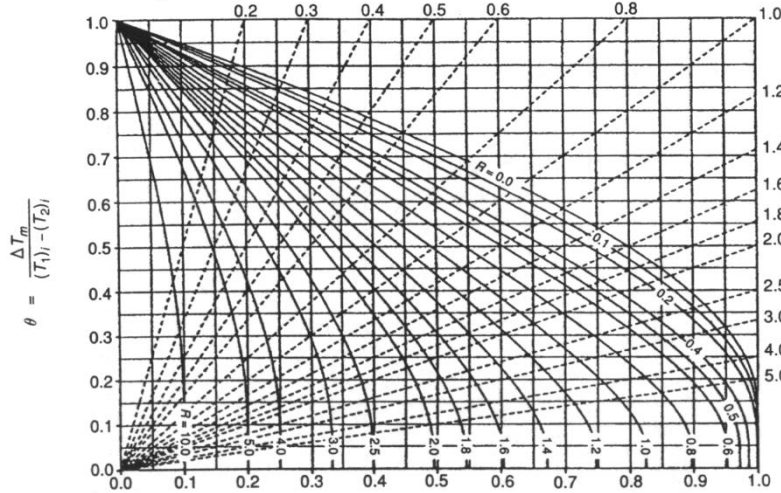
$$\rightarrow P \text{ Thermal Effectiveness} = \frac{(T_2)_o - (T_2)_i}{(T_1)_i - (T_2)_i}$$

$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

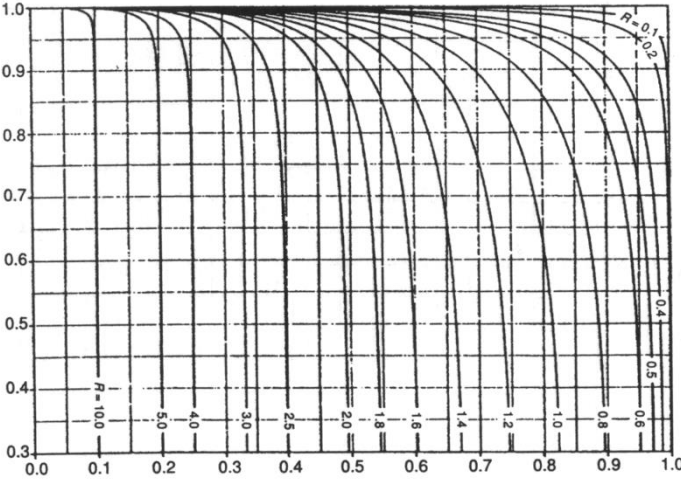
T_1 and T_2 are not interchangeable



→ $NTU_2 = AU/C_2$



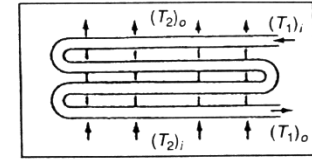
$$F \text{ Correction Factor} = \frac{\Delta T_m}{\Delta T_{lm}}$$



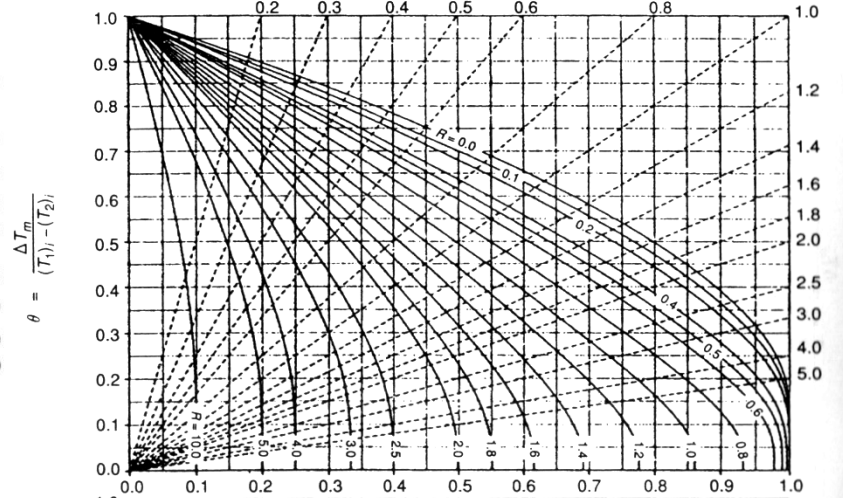
→ $P \text{ Thermal Effectiveness} = \frac{(T_2)_o - (T_2)_i}{(T_1)_i - (T_2)_i}$

$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

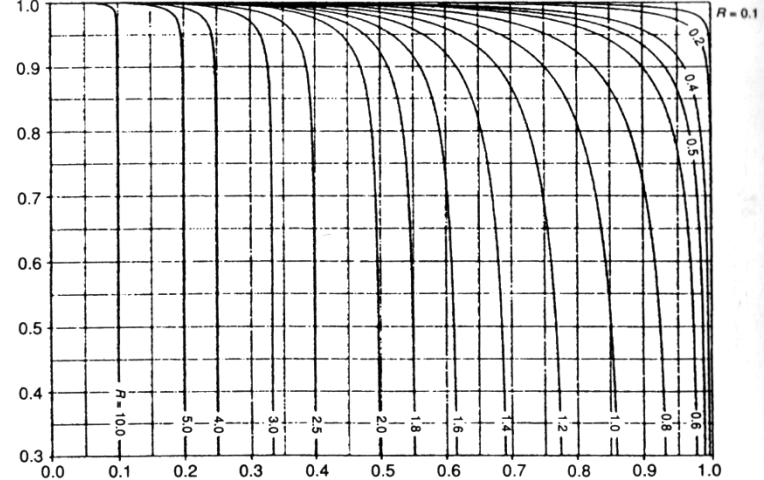
T_1 and T_2 are not interchangeable



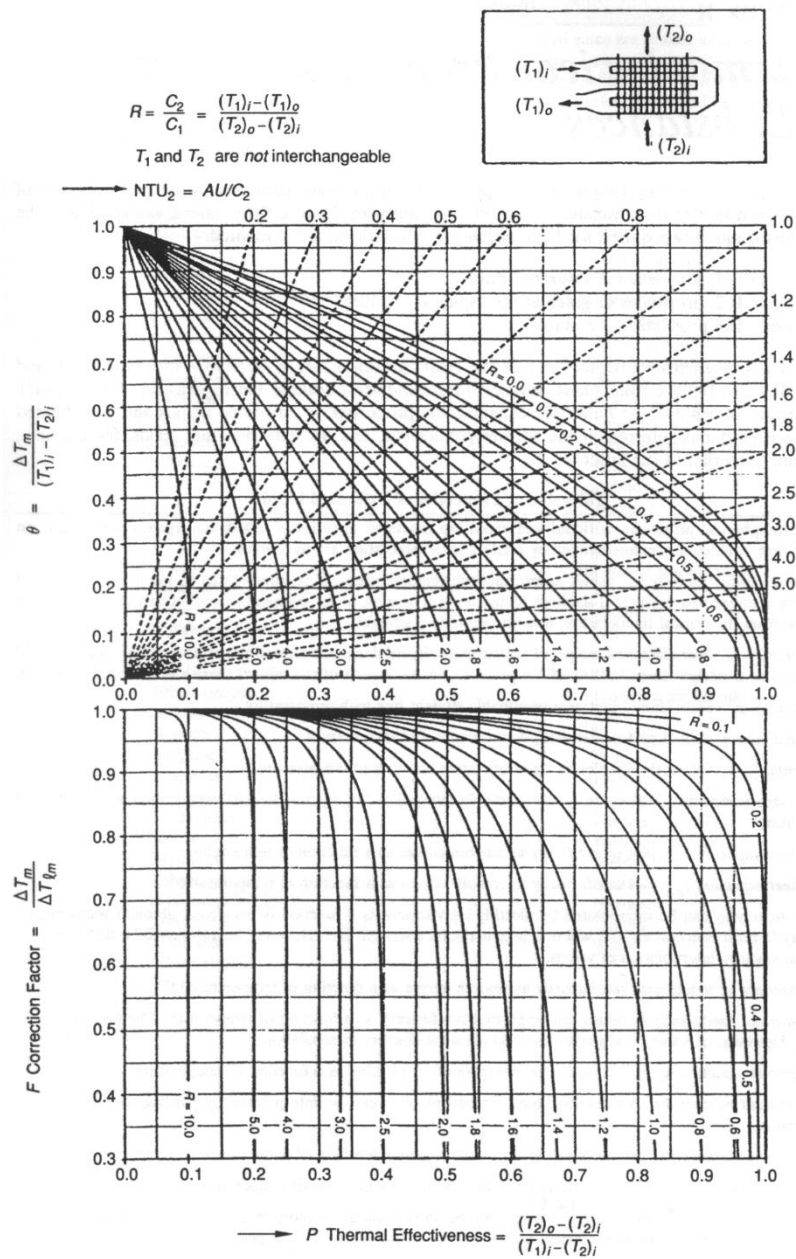
→ $NTU_2 = AU/C_2$



$$F \text{ Correction Factor} = \frac{\Delta T_m}{\Delta T_{lm}}$$



→ $P \text{ Thermal Effectiveness} = \frac{(T_2)_o - (T_2)_i}{(T_1)_i - (T_2)_i}$



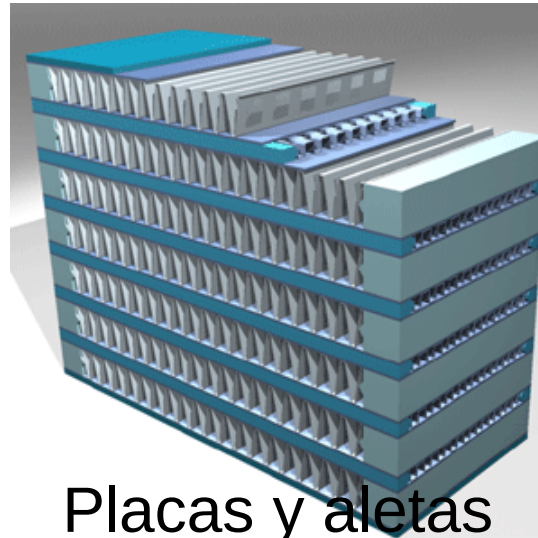




Placas con juntas



Placas soldadas



Placas y aletas



Doble tubo



Tubo carcasa



Placas empacadas

Refrigerados por aire

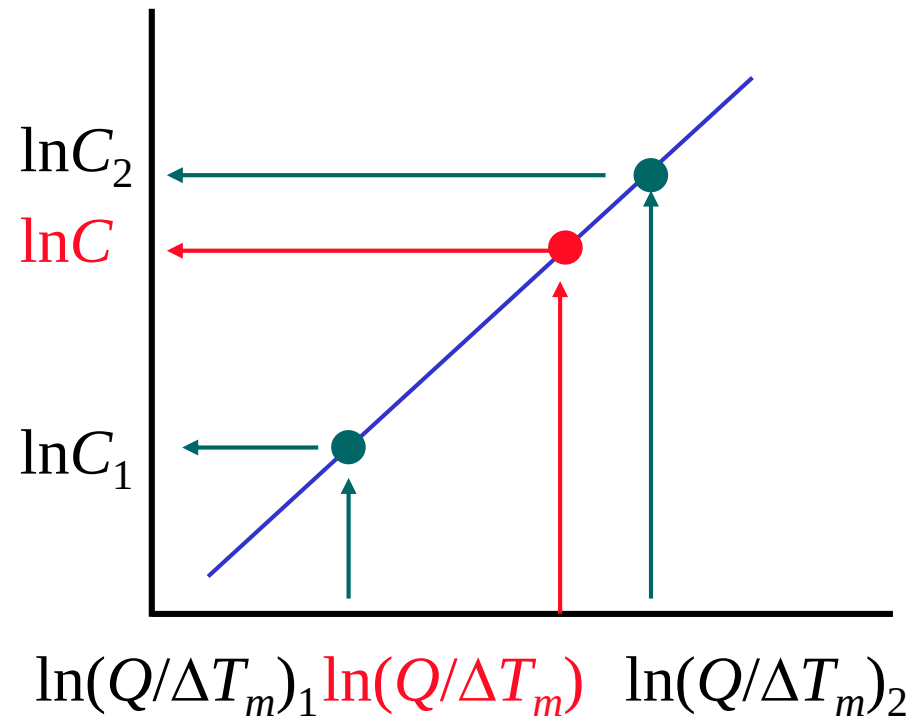


Tipo de intercambiador	Maxima presión (absoluta)	Rango de temperaturas	Limitación del fluido	Rango normal de tamaños para una unidad	Cualidades especiales
Refrigerado por aire	Alta en la zona de proceso (500 bar).	High temperature possible on process side (up to 600°C).	Restringido sólo por el material de construcción	5 a 350 m ² (tubo desnudo) (por matriz – se usan normalmente multiples matrices).	Sistema de evacuación del calor. Necesita ventiladores. Los tubos normalmente están aleteados.
Placas (empaquetadas)	16 bar	Sobre 200°C	Sólo para líquidos compatibles con la carcasa	1 a 10 m ²	Construcción modular. Limpieza dificultosa
Doble tubo	300 bar (carcasa) (posibilidad de mayores presiones en diseños especiales). 1400 bar (tubos).	-100 a 600°C (mayores con materials especiales).	Restringido sólo por el material de construcción	0.25 a 200 m ² por unidad – se usan normalmente múltiples unidades	Alta eficiencia térmica, construcción modular estándar.
Placas (juntas)	25 bar algunos modelos especiales sobre 40 bar.	– 25 to 175°C (–40 a 200°C posible en modelos especiales)	No usual para gases. Limitación en las juntas	1 a 2500 m ²	Construcción modular. Normalmente el más económico si cumple las condiciones
Placas-aletas	100 bar (aluminio). 200 bar (acero).	–273 to + 150°C (aluminio) (sobre 600°C en acero).	Bajo ensuciamiento	<9m ³ volumen	Existen muy pequeños. Se pueden usar con multiples fluidos. Gran relación area - volumen
Tubo carcasa	300 bar (carcasa). 1400 bar (tubos).	–25 a 600°C (mayors y menores con materials especiales).	Restringido sólo por el material de construcción	10 a 1000 m ² (por carcasa – se pueden usar multiples carcasas).	Muy adaptable, se puede usar en casi todas las aplicaciones
Placas (soldadas)	60 bar	Por encima de 650°C	Restringido sólo por el material de construcción. No se aconseja en fluidos sucios	>1000 m ²	La presión diferencial debería ser menor de 30 bar. Se deben tener en cuenta las expansiones diferenciales

Depende del tipo de fluidos en el intercambiador

Aproximación $F=0.9$ multitubulares, $F=0.7$ flujo cruzado

$$C = \exp \left(\ln C_1 + \ln (C_1/C_2) \frac{\ln \left[(Q/\Delta T_m)/(Q/\Delta T_m)_1 \right]}{\ln \left[(Q/\Delta T_m)_1/(Q/\Delta T_m)_2 \right]} \right)$$



Para cada tipo de intercambiador y UA los constructores han tabulado el coste y el coeficiente global en función de los fluidos que intercambian

$$Q/\Delta T_m = UA$$

Fluido Frío Fluido Caliente U
 C

$Q/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
1 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W.K))	55 5.70	93 5.02	120 5.51	102 4.93	99 4.96	63 5.50	107 4.87	100 4.95	86 5.11
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W.K))	93 5.02	300 4.18	350 4.81	420 4.03	375 4.09	120 4.76	530 3.95	388 4.07	240 4.28
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W.K))	120 5.51	350 4.81	400 6.25	600 4.56	450 4.38	200 5.50	600 4.56	400 4.82	300 4.81
	Treated Cooling Water	U (W/m ² K) C (£/(W.K))	105 4.89	484 3.98	600 4.56	938 3.77	714 3.85	142 4.59	1607 3.61	764 3.83	345 4.12
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W.K))	99 4.96	375 4.09	450 4.38	600 3.91	500 3.97	130 4.67	818 3.81	524 3.95	286 4.20
	High Viscosity Liquid	U (W/m ² K) C (£/(W.K))	68 5.39	138 4.61	200 5.50	161 4.46	153 4.51	82 5.16	173 4.42	155 4.50	214 4.33
	Boiling Water	U (W/m ² K) C (£/(W.K))	105 4.89	467 3.99	550 4.91	875 3.79	677 3.87	140 4.60	1432 3.64	722 3.85	336 4.13
	Boiling Organic Liquid	U (W/m ² K) C (£/(W.K))	99 4.96	375 4.09	450 4.38	600 3.91	500 3.97	130 4.67	818 3.81	524 3.95	286 4.20

Coste (tubo-carcasa) I

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
1 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 5.70	93 5.02	120 5.51	102 4.93	99 4.96	63 5.50	107 4.87	100 4.95	86 5.11
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 5.02	300 4.18	350 4.81	429 4.03	375 4.09	120 4.76	530 3.95	388 4.07	240 4.28
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 5.51	350 4.81	400 6.25	600 4.56	450 4.38	200 5.50	600 4.56	400 4.82	300 4.81
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 4.89	484 3.98	600 4.56	938 3.77	714 3.85	142 4.59	1607 3.61	764 3.83	345 4.12
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 4.96	375 4.09	450 4.38	600 3.91	500 3.97	130 4.67	818 3.81	524 3.95	286 4.20
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 5.39	138 4.61	200 5.50	161 4.46	153 4.51	82 5.16	173 4.42	155 4.50	214 4.33
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 4.89	467 3.99	550 4.91	875 3.79	677 3.87	140 4.60	1432 3.64	722 3.85	336 4.13
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 4.96	375 4.09	450 4.38	600 3.91	500 3.97	130 4.67	818 3.81	524 3.95	286 4.20
5 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 2.11	93 1.63	120 2.26	102 1.58	99 1.59	63 1.95	107 1.55	100 1.59	86 1.68
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 1.63	300 1.11	350 1.89	429 1.02	375 1.05	120 1.49	530 0.98	388 1.05	240 1.18
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 2.26	350 1.89	400 2.25	600 1.10	450 1.46	200 1.93	600 1.10	400 1.45	300 1.45
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 1.56	484 1.00	600 1.10	938 0.88	720 0.91	142 1.41	1607 0.83	764 0.90	345 1.07
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 1.59	375 1.05	450 1.46	600 0.95	500 0.99	130 1.46	818 0.89	524 0.98	286 1.13
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 1.86	138 1.43	200 1.93	161 1.36	153 1.38	82 1.71	173 1.32	155 1.37	124 1.48
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 1.56	467 1.00	550 1.20	875 0.88	677 0.93	140 1.42	1432 0.84	722 0.91	336 1.08
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 1.59	375 1.05	450 1.46	600 0.95	500 0.99	130 1.46	818 0.89	524 0.98	286 1.13

Coste (tubo-carcasa) II

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
30 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 1.11	93 0.76	120 1.06	102 0.73	99 0.74	63 0.99	107 0.71	100 0.73	86 0.80
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 0.76	300 0.37	350 0.62	500 0.28	375 0.33	120 0.63	530 0.28	388 0.32	240 0.42
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 1.06	350 0.62	400 0.94	600 0.40	450 0.53	200 0.73	600 0.40	400 0.62	300 0.62
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 0.71	484 0.29	600 0.40	938 0.23	714 0.25	142 0.56	1607 0.19	764 0.24	345 0.34
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.38	130 0.59	818 0.24	524 0.28	286 0.38
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 0.94	138 0.57	200 0.73	161 0.52	153 0.53	82 0.83	173 0.50	155 0.53	124 0.60
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 0.71	467 0.29	550 0.49	875 0.23	677 0.25	140 0.56	1432 0.20	722 0.25	336 0.35
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.28	130 0.59	818 0.24	524 0.28	286 0.38
100 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 0.95	93 0.58	120 0.93	102 0.54	99 0.55	63 0.83	107 0.52	100 0.55	86 0.62
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 0.58	300 0.23	350 0.35	429 0.18	375 0.20	120 0.47	530 0.16	388 0.19	240 0.27
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 0.93	350 0.35	400 0.58	600 0.24	450 0.28	200 0.64	600 0.24	400 0.32	300 0.39
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 0.52	484 0.17	600 0.24	938 0.116	714 0.134	142 0.41	1607 0.086	764 0.129	345 0.21
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.55	375 0.20	450 0.28	609 0.145	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 0.77	138 0.42	200 0.64	161 0.37	153 0.38	82 0.65	173 0.35	155 0.38	124 0.45
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 0.52	467 0.168	550 0.26	875 0.121	677 0.137	140 0.41	1432 0.091	722 0.133	336 0.22
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.55	375 0.20	450 0.28	600 0.146	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
1 000 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 0.93	93 0.55	120 0.93	102 0.50	99 0.52	63 0.81	107 0.48	100 0.51	86 0.60
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 0.55	300 0.171	350 0.32	500 0.103	375 0.137	120 0.43	530 0.098	388 0.132	240 0.21
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 0.93	350 0.32	400 0.58	600 0.19	450 0.25	200 0.56	600 0.19	400 0.28	300 0.37
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 0.49	484 0.106	600 0.19	938 0.058	714 0.074	142 0.36	1607 0.037	764 0.069	345 0.148
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.52	375 0.137	450 0.25	600 0.087	500 0.103	130 0.39	818 0.065	524 0.099	286 0.179
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 0.75	138 0.37	200 0.56	161 0.32	153 0.34	82 0.62	173 0.30	155 0.33	124 0.41
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 0.49	467 0.11	550 0.20	875 0.061	677 0.077	140 0.37	1432 0.04	722 0.073	336 0.152
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.52	375 0.137	450 0.25	600 0.087	500 0.103	130 0.39	818 0.065	524 0.098	280 0.179

Note: The C values given in the above table are based on the assumption that the exchangers are constructed from carbon steel with a design pressure appropriate to the working pressure. The U -values take account of 'normal' fouling and are based on the outside surface area of the tubes. The costs are based on BEM designs with a minimum shell length to shell diameter ratio of 12. The maximum working pressure at which these approximate costs are valid is 20 bar, except for the cases where high pressure gas is one of the fluids. Similarly, the maximum temperature at which these approximate costs are valid is 400°C. Further useful information on the effect of design parameters on costs is given by G.P. Purohit ("Estimating costs of shell-and-tube heat exchangers", *Chemical Engineering*, pp. 56-67, Aug. 22nd, 1983). Construction from stainless steel costs approximately 1.5 times more at pressures up to 30 bar and approximately 3 times more at a pressure of 150 bar.

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Hydrocarbon Liquid	High Viscosity Hydrocarbon Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
1 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 4.8	95 3.8	125 2.9	105 3.8	100 3.8	65 4.7	110 2.8	100 3.8	85 3.9
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	95 3.4	300 2.5	350 2.9	430 2.5	375 5	120 2.6	530 2.5	390 2.5	240 2.5
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 2.9	350 2.9	400 2.9	500 2.9	400 2.9	150 2.9	600 2.9	420 2.9	350 2.9
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 2.8	484 2.5	500 2.9	940 2.5	715 2.5	145 2.5	1610 2.5	765 3.9	345 2.5
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	100 2.7	375 2.5	425 2.9	600 2.5	500 2.5	130 2.5	820 2.5	525 2.5	290 2.5
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	70 4.7	140 2.5	175 2.9	160 2.5	155 2.5	85 3.9	175 2.5	155 2.5	125 2.5
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 3.9	470 2.5	550 2.9	875 2.5	670 2.5	140 2.5	1435 2.5	725 2.5	340 2.5
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	100 4	375 2.5	430 2.9	600 2.5	500 2.5	130 2.5	820 2.5	525 2.5	285 2.5
5 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 2.16	95 1.26	125 1.1	105 1.23	100 1.22	65 1.84	110 1.23	100 1.22	85 3.5
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	95 1.5	300 0.86	350 1	430 0.76	375 0.8	120 1.16	530 0.6	390 0.8	240 0.94
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 2.05	350 1.1	400 1.1	500 1	400 1.1	150 1.3	600 1	420 1.1	350 1.2
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 1.4	484 0.75	500 1	940 0.5	715 0.72	145 1	1610 0.4	765 0.72	345 0.9
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	100 1.45	375 0.8	425 1.05	600 0.8	500 0.9	130 1.1	820 0.5	525 0.9	290 1.2
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	70 2.4	140 1.66	175 1.2	160 0.95	155 1	85 2.7	175 0.82	155 1.6	125 1.95
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 1.4	470 0.9	550 1	875 0.5	670 0.6	140 1.1	1435 0.4	725 0.73	340 0.9
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	100 1.45	375 0.8	430 1.05	600 0.8	500 0.9	130 1.28	820 0.7	525 1	285 1.1

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Hydrocarbon Liquid	High Viscosity Hydrocarbon Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
30 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 2.16	95 1.26	125 1.1	105 1.23	100 1.22	65 1.84	110 1.23	100 1.22	85 3.5
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	95 1.5	300 0.43	350 0.35	430 0.27	375 0.33	120 1.16	530 0.23	390 3.07	240 0.42
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 2.05	350 0.4	400 0.36	500 0.27	400 0.33	150 1.5	600 0.24	420 0.3	350 0.35
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 1.4	484 0.31	500 0.27	940 0.19	715 0.21	145 1	1610 0.17	765 0.2	345 0.37
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	100 1.45	375 0.33	425 0.28	600 0.17	500 0.19	130 1.1	820 0.18	525 0.27	290 0.41
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	70 2.4	140 1.8	175 0.55	160 0.62	155 0.65	85 2.7	175 1.66	155 0.68	125 1.95
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 1.4	470 0.26	550 0.25	875 0.17	670 0.21	140 1.1	1435 0.17	725 0.2	340 0.36
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	100 1.45	375 0.33	430 0.31	600 0.22	500 0.24	130 1.28	820 0.2	525 0.23	285 0.46
100 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 2.16	95 1.26	125 1.1	105 1.23	100 1.22	65 1.84	110 1.23	100 1.22	85 3.5
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	95 1.5	300 0.43	350 0.35	430 0.27	375 0.33	120 1.16	530 0.18	390 3.07	240 0.42
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 2.05	350 0.4	400 0.36	500 0.27	400 0.33	150 1.5	600 0.2	420 0.3	350 0.35
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 1.4	484 0.31	500 0.22	940 0.13	715 0.14	145 1	1610 0.11	765 0.2	345 0.37
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	100 1.45	375 0.33	425 0.28	600 0.17	500 0.19	130 1.1	820 0.14	525 0.19	290 0.41
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	70 2.4	140 1.8	175 0.55	160 0.62	155 0.65	85 2.7	175 1.66	155 0.68	125 1.95
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 1.4	470 0.26	550 0.25	875 0.14	670 0.16	140 1.1	1435 0.11	725 0.14	340 0.36
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	100 1.45	375 0.33	430 0.31	600 0.17	500 0.23	130 1.28	820 0.12	525 0.23	285 0.46

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Hydrocarbon Liquid	High Viscosity Hydrocarbon Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
1 000 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 2.16	95 1.26	125 1.1	105 1.23	100 1.22	65 1.84	110 1.23	100 1.22	85 3.5
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	95 1.5	300 0.43	350 0.35	430 0.27	375 0.33	120 1.16	530 0.18	390 3.07	240 0.42
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 2.05	350 0.4	400 0.36	500 0.27	400 0.33	150 1.5	600 0.2	420 0.3	350 0.35
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 1.4	484 0.31	500 0.22	940 0.13	715 0.14	145 1	1610 0.11	765 0.2	345 0.37
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	100 1.45	375 0.33	425 0.28	600 0.17	500 0.19	130 1.1	820 0.14	525 0.19	290 0.41
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	70 2.4	140 1.8	175 0.55	160 0.62	155 0.65	85 2.7	175 1.66	155 0.68	125 1.95
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 1.4	470 0.26	550 0.25	875 0.14	670 0.16	140 1.1	1435 0.11	725 0.14	340 0.36
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	100 1.45	375 0.33	430 0.31	600 0.17	500 0.23	130 1.28	820 0.12	525 0.23	285 0.46

Coste (placas con juntas)

$\dot{Q}/\Delta T$ (W/K)	Cold Side Liquid	Parameter	Hot Side Liquid		
			Water	Aqueous Inorganic Liquid	Organic Liquid
5 000	Water	U (W/m ² K) C (£/(W/K))	5500 0.160	3500 0.130 to 0.170	1750 0.140 to 0.190
	Aqueous Inorganic Liquid	U (W/m ² K) C (£/(W/K))	3500 0.130 to 0.170	1750 0.190	1250 0.160 to 0.210
	Organic Liquid	U (W/m ² K) C (£/(W/K))	1750 0.140 to 0.190	1250 0.160 to 0.210	1150 0.220
10 000	Water	U (W/m ² K) C (£/(W/K))	5500 0.1	3500 0.1	1850 0.11
	Aqueous Inorganic Liquid	U (W/m ² K) C (£/(W/K))	3500 0.1	2200 0.1	1400 0.13
	Organic Liquid	U (W/m ² K) C (£/(W/K))	1850 0.11	1400 0.13	1150 0.15
50 000	Water	U (W/m ² K) C (£/(W/K))	5500 0.025	3500 0.025 to 0.035	1850 0.035
	Aqueous Inorganic Liquid	U (W/m ² K) C (£/(W/K))	3500 0.025 to 0.035	3500 0.035	1400 0.055
	Organic Liquid	U (W/m ² K) C (£/(W/K))	1850 0.035	1400 0.055	1150 0.065
100 000	Water	U (W/m ² K) C (£/(W/K))	5500 0.025	3500 0.035	1850 0.045
	Aqueous Inorganic Liquid	U (W/m ² K) C (£/(W/K))	3500 0.035	3500 0.035	1400 0.055
	Organic Liquid	U (W/m ² K) C (£/(W/K))	1850 0.045	1400 0.055	1150 0.065
500 000	Water	U (W/m ² K) C (£/(W/K))	5500 0.015	3500 0.016	1850 0.020
	Aqueous Inorganic Liquid	U (W/m ² K) C (£/(W/K))	3500 0.016	3500 0.020	1400 0.035
	Organic Liquid	U (W/m ² K) C (£/(W/K))	1850 0.020	1400 0.035	1150 0.045
1 000 000	Water	U (W/m ² K) C (£/(W/K))	5500 0.015	3500 0.016	1850 0.025
	Water	U (W/m ² K) C (£/(W/K))	5500 0.015	3500 0.016	1850 0.025
20 000 000	Water	U (W/m ² K) C (£/(W/K))	5500 0.015	3500 0.016	1850 0.025

Notes: 1. Cost figures refer to stainless steel plates and operating pressures of 10 bar.

2. If titanium plate is used costs will be 1.7 to 2.0 times higher.

3. If pressure is increased to 25 bar costs will be 1.3 to 1.5 times higher.

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Hydrocarbon Liquid	High Viscosity Hydrocarbon Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
5 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	163 3.10	217 3.10	NA	NA	264 3.10	NUS	NUS	270 3.10	NUS
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	217 3.10	325 3.10	NA	NA	377 3.10	NUS	NUS	402 3.10	NUS
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	315 3.10	491 3.10	NA	NA	NUS	NUS	NUS	NUS	NUS
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NA	NUS	NUS	NUS	NUS	NUS
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NUS	NUS	NUS	NUS	NUS	NUS
	Boiling water	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Boiling Hydrocarbon	U (W/m ² K) C (£/(W/K))	270 3.10	402 3.10	NA	NA	453 3.10	NUS	NA	530 3.10	NUS
10 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	163 1.57	217 1.55	NA	NA	264 1.55	NUS	NUS	270 1.55	NUS
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	217 1.55	325 1.55	NA	NA	377 1.55	NUS	NUS	402 1.55	NUS
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	315 1.55	491 1.55	NA	NA	NUS	NUS	NUS	NUS	NUS
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NA	NUS	NUS	NUS	NUS	NUS
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NUS	NUS	NUS	NUS	NUS	NUS
	Boiling water	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Boiling Hydrocarbon	U (W/m ² K) C (£/(W/K))	270 1.55	402 1.55	NA	NA	453 1.55	NUS	NA	530 1.55	NUS

(Key: NA - Not Applicable; NUS - Not Common)

Note: The C values given in the above table are based on the assumption that the heat exchangers are made of brazed aluminium and are designed for pressures less than 25 bar and temperatures below 340 K. If the pressure is increased to 80 bar, the cost will increase approximately 1.25 times. If the heat exchangers are made from stainless steel, the cost will be about 3 times higher, and if made from titanium the cost will be about 5 times higher.

Coste (placas y aletas) II

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Hydrocarbon Liquid	High Viscosity Hydrocarbon Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
30 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	163 0.677	217 0.607	NA	NA	264 0.574	NUS	NUS	270 0.579	NUS
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	217 0.607	325 0.551	NA	NA	377 0.537	NUS	NUS	402 0.532	NUS
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	315 0.560	491 0.513	NA	NA	NUS	NUS	NUS	NUS	NUS
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NA	NUS	NUS	NUS	NUS	NUS
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NUS	NUS	NUS	NUS	NUS	NUS
	Boiling water	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Boiling Hydrocarbon	U (W/m ² K) C (£/(W/K))	270 0.579	402 0.532	NA	NA	453 0.527	NUS	NA	530 0.518	NUS
100 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	163 0.336	217 0.301	NA	NA	264 0.280	NUS	NUS	270 0.273	NUS
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	217 0.301	325 0.245	NA	NA	377 0.231	NUS	NUS	402 0.227	NUS
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	315 0.250	491 0.210	NA	NA	NUS	NUS	NUS	NUS	NUS
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NA	NUS	NUS	NUS	NUS	NUS
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NUS	NUS	NUS	NUS	NUS	NUS
	Boiling water	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Boiling Hydrocarbon	U (W/m ² K) C (£/(W/K))	270 0.273	402 0.227	NA	NA	453 0.216	NUS	NA	530 0.205	NUS
1 000 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	163 0.201	217 0.168	NA	NA	264 0.152	NUS	NUS	270 0.151	NUS
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	217 0.168	325 0.139	NA	NA	377 0.128	NUS	NUS	402 0.126	NUS
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	315 0.140	491 0.115	NA	NA	NUS	NUS	NUS	NUS	NUS
	Low Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NA	NUS	NUS	NUS	NUS	NUS
	High Viscosity Hydrocarbon Liquid	U (W/m ² K) C (£/(W/K))	NUS	NUS	NA	NUS	NUS	NUS	NUS	NUS	NUS
	Boiling water	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Boiling Hydrocarbon	U (W/m ² K) C (£/(W/K))	270 0.151	402 0.126	NA	NA	453 0.119	NUS	NA	530 0.110	NUS

Cold Side Fluid - Air		Hot-Side Fluid								
$\dot{Q}/\Delta T$ (W/K)	Parameter	Low Pressure Gas 1 bar	Medium Pressure Gas 20 bar	High Pressure Gas 150 bar	Treated Water	Low Viscosity Hydrocarbon Liquid	Medium Viscosity Hydrocarbon Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Gas
1 000	U (W/m ² K) C (£/(W/K))	200 13.68	350 12.21	450 14.53	700 11.81	400 12.60	250 13.19	800 11.44	350 12.49	300 14.38
5 000	U (W/m ² K) C (£/(W/K))	200 4.35	400 3.48	550 3.51	700 2.98	400 3.58	250 3.96	800 2.59	400 3.49	350 3.64
30 000	U (W/m ² K) C (£/(W/K))	230 1.64	450 1.17	600 1.12	750 0.89	450 1.06	300 1.45	800 0.81	500 1.08	400 1.35
100 000	U (W/m ² K) C (£/(W/K))	230 1.15	450 0.78	600 0.59	750 0.45	450 0.58	300 0.92	800 0.43	500 0.56	400 0.80
1 000 000	U (W/m ² K) C (£/(W/K))	230 1.11	450 0.53	600 0.46	750 0.28	450 0.48	300 0.76	800 0.31	500 0.40	400 0.61

Note: Overall heat transfer coefficient U is referred to outside bare surface area.

U and C values are based on the use of 1 inch (24.4 mm) outside diameter carbon steel tubes with 2.25 inch (57.2 mm) outside diameter with spiral wound, embedded aluminium fins at 11 fins per inch (433 fins per metre) and on the use of carbon steel plug type headers. The cost calculations were done for forced draught operations with horizontal bundles of 4 to 6 rows and 2 to 12 m length (the number of rows and lengths were chosen appropriate to the respective duties). For stainless steel, the cost would increase by 1.5 to 2.5 times. It should be noted that a large number of factors affect the cost of air-cooled exchangers, including fin type, tube lengths, number of tube rows, header type, number of fans, noise levels, structure finish and extent of walkways. Information about the effect of these factors on the cost of air-cooled heat exchangers is given by C. North ("Costing of air coolers", Heat Exchanger Design Handbook, pp. 4.3.3-1 to 4.8.3-3, Hemisphere Publishing Corp., New York, 1983). The above table gives C -values for typical choices of these parameters but the considerable variability should be borne in mind when using the table.

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot-side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
1 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	78 8.1	106 8.0	NA	220 4.9	153 5.7	122 5.9	243 4.7	225 4.9	190 5.8
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	103 8.1	256 7	NA	381 4.9	308 4.9	202 5.7	1021 4.0	708 3.7	600 4.36
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	158 5.6	349 4.9	NA	1328 3.7	380 4.1	228 4.9	1750 3.5	511 3.9	400 4.98
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	143 5.1	297 5.6	NA	343 4.1	534 3.9	283 4.4	1085 3.8	701 3.7	600 4.36
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	103 6.1	188 5.9	NA	215 4.8	271 4.6	167 5.1	363 3.9	311 4.1	220 4.91
	Boiling Water	U (W/m ² K) C (£/(W/K))	170 5.20	350 4.90	NA	1400 3.60	500 3.90	250 4.32	1800 3.50	800 3.28	500 4.36
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	150 5.50	300 5.6	NA	1000 3.80	400 4.98	200 4.46	1200 3.70	600 4.36	400 4.98
5 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	77 4.02	114 3.4	NA	220 2.8	168 2.74	119 2.8	243 4.4	226 1.92	190 2.28
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	108 3.58	616 1.5	NA	1223 1.18	1110 1.22	357 1.98	1400 1.02	787 1.3	440 1.98
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	215 2.54	1187 1.22	NA	4252 0.74	1551 1.0	454 1.52	3880 0.7	1272 0.9	600 1.22
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	140 2.94	975 1.22	NA	1650 0.82	1692 0.78	522 1.22	2676 0.78	1281 0.92	600 1.22
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	131 2.8	357 0.9	NA	411 1.3	400 1.8	242 2.04	621 1.26	434 1.36	300 1.66
	Boiling Water	U (W/m ² K) C (£/(W/K))	195 2.8	1000 1.28	NA	4500 0.78	2000 0.58	600 1.04	4000 0.75	1200 1.28	550 1.35
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	155 2.65	800 1.48	NA	2000 0.79	1500 0.80	500 1.25	2500 0.78	900 1.38	450 1.52

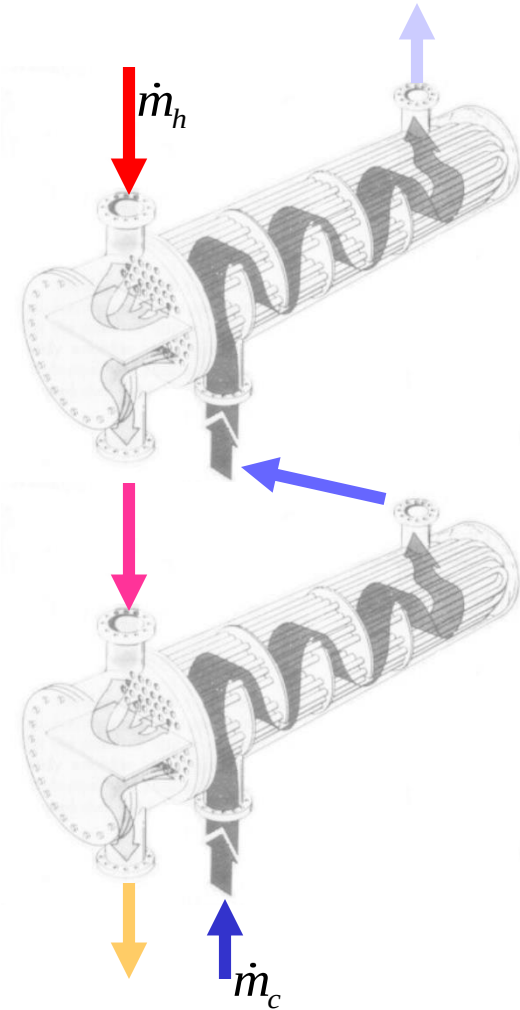
(Key: NA - Not Applicable; NUS - Not Common)

Note: The C-values given in the above table are based on the assumption that the exchangers are constructed from stainless steel type 316 with a design pressure appropriate to the working pressure. The costs are valid up to 600°C. If the external vessel is constructed from carbon steel, the costs would be 20-40 per cent less depending on size and pressure.

Coste (Placas soldados)II

$\dot{Q}/\Delta T$ (W/K)		Cold Side Fluid	Parameter	Hot-side Fluid							
				Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon
30 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	83 2.33	133 1.8	NA	238 1.67	176 1.96	134 1.95	246 0.934	191 1.01	170 1.96
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	129 2.1	503 0.733	NA	1262 0.43	819 0.54	774 0.61	1384 0.494	850 0.634	440 0.52
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	210 1.53	1068 0.42	NA	9100 0.147	4268 0.26	2067 0.26	5465 0.18	1518 0.274	600 0.55
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	141 1.82	1060 0.37	NA	3570 0.21	3763 0.194	1075 0.367	3652 0.214	1366 0.297	600 0.55
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	131 1.9	298 0.87	NA	484 0.52	413 0.584	292 0.824	1626 0.287	403 0.54	300 0.87
	Boiling Water	U (W/m ² K) C (£/(W/K))	220 1.51	1100 0.42	NA	9000 0.15	4500 0.25	2000 0.26	5500 0.18	1500 0.28	600 0.55
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	160 1.65	900 0.39	NA	5000 0.19	3500 0.20	1200 0.37	4000 0.26	1000 0.37	500 0.53
100 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	65 3.84	141 1.33	NA	230 1.42	167 1.74	141 2.0	246 1.06	190 0.92	170 1.75
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	137 1.38	534 0.462	NA	1354 0.303	940 0.34	776 0.41	1342 0.238	834 0.307	440 0.51
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	223 1.37	1112 0.28	NA	9420 0.062	4858 0.108	2106 0.13	5071 0.085	1518 0.186	600 0.40
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	157 1.71	722 0.36	NA	4140 0.118	3794 0.101	1277 0.19	3712 0.108	1367 0.161	600 0.40
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	133 1.67	355 0.592	NA	530 0.42	415 0.538	302 0.515	1617 0.158	452 0.389	300 0.52
	Boiling Water	U (W/m ² K) C (£/(W/K))	220 1.37	1100 0.28	NA	9000 0.063	4500 0.13	2000 0.15	5500 0.09	1500 0.17	600 0.40
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	160 1.71	900 0.32	NA	5000 0.12	3500 0.14	1200 0.19	4000 0.13	1000 0.30	500 0.43
1 000 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	65 3.84	141 1.33	NA	230 1.42	167 1.74	141 2.00	246 1.06	190 0.92	170 1.75
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	137 1.38	504 0.41	NA	1334 0.22	989 0.242	787 0.282	1418 0.127	826 0.241	440 0.43
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	223 1.37	1173 0.22	NA	9050 0.033	5061 0.045	2122 0.068	5460 0.038	1561 0.095	600 0.20
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	157 1.71	927 0.23	NA	4880 0.04	4104 0.0505	1148 0.123	3828 0.048	1379 0.1	600 0.20
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	133 1.67	323 0.49	NA	510 0.21	553 0.2	312 0.321	1714 0.081	434 0.241	300 0.33
	Boiling Water	U (W/m ² K) C (£/(W/K))	220 1.37	1100 0.12	NA	9000 0.033	4500 0.05	2000 0.07	5500 0.04	1500 0.10	600 0.20
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	160 1.71	900 0.14	NA	5000 0.05	3500 0.051	1200 0.13	4000 0.041	1000 0.12	500 0.22

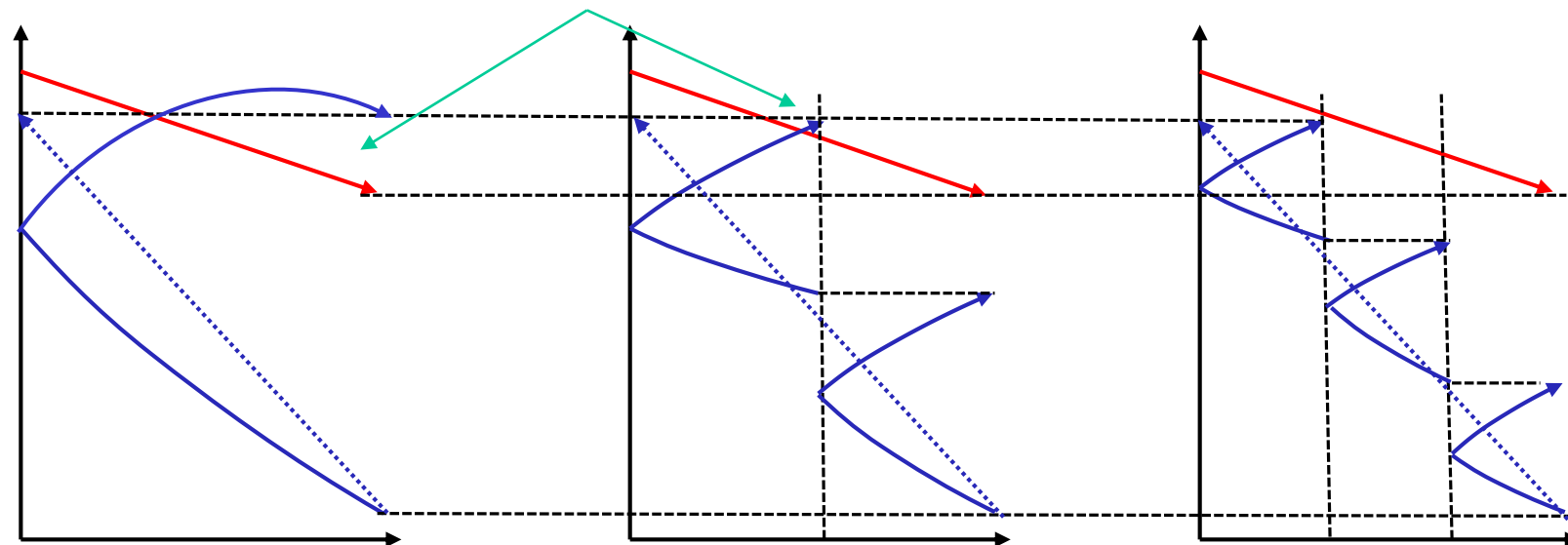
Ambos en serie (multicarcasas)



Cuando la potencia del intercambiador es muy alta debido fundamentalmente a un **gran salto de temperaturas** se utilizan combinaciones en serie



Zona no eficiente



1 paso por la carcasa

2 pasos por la carcasa

3 pasos por la carcasa

A medida que aumentan el número de carcassas el perfil térmico se parece más al de contracorriente

$$\varepsilon_1 = \varepsilon_2 = \varepsilon_3$$

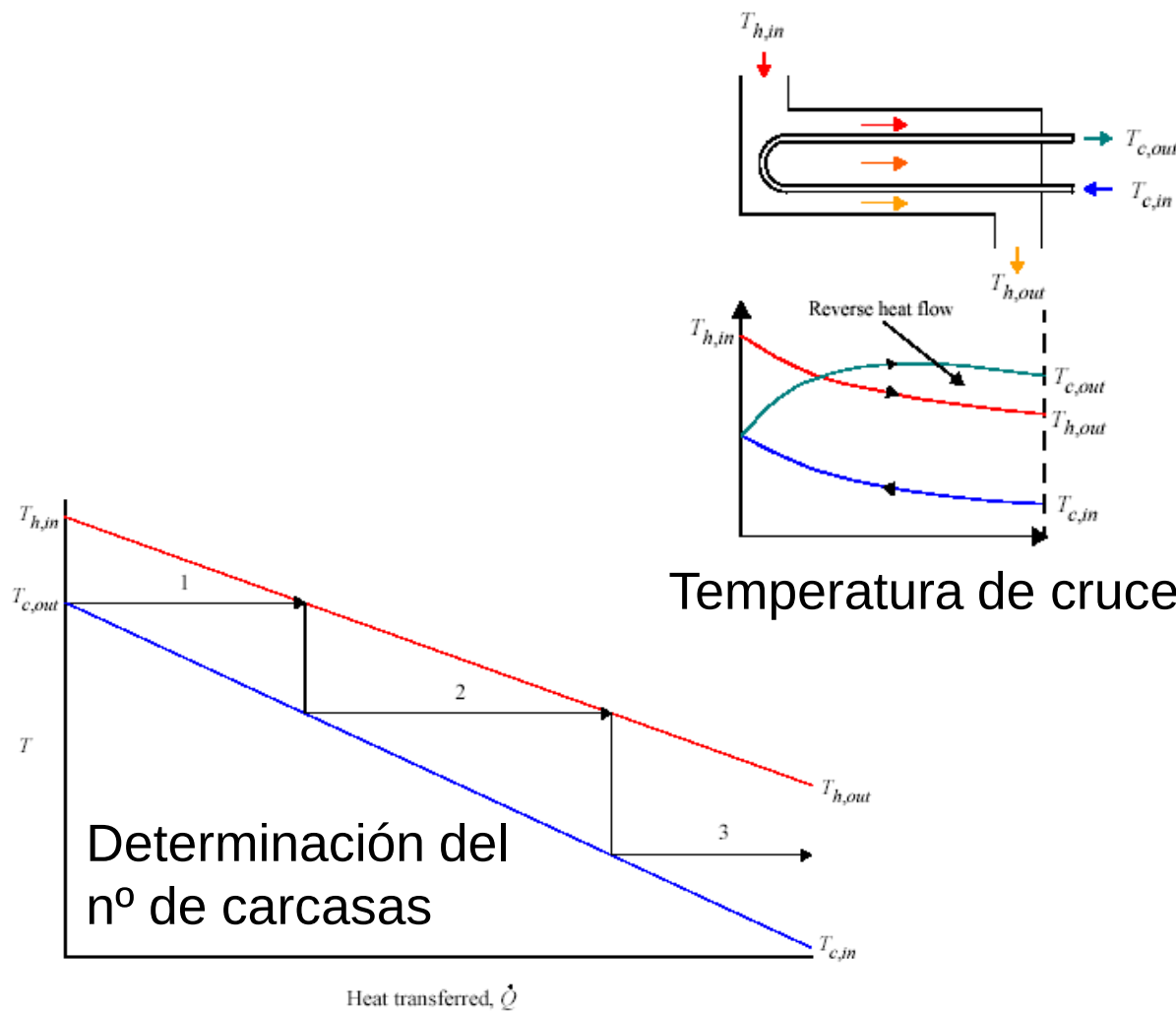
$$P_1 = P_2 = P_3$$

$$A_1 > A_2 > A_3$$

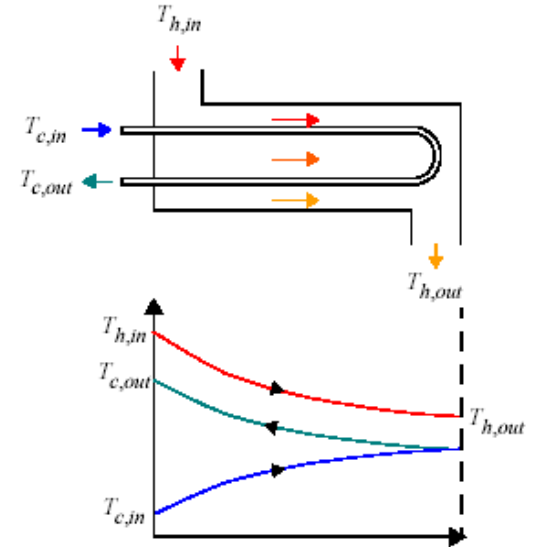
$$F_1 < F_2 < F_3$$

F es una especie de eficiencia económica

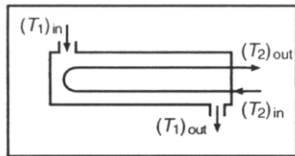
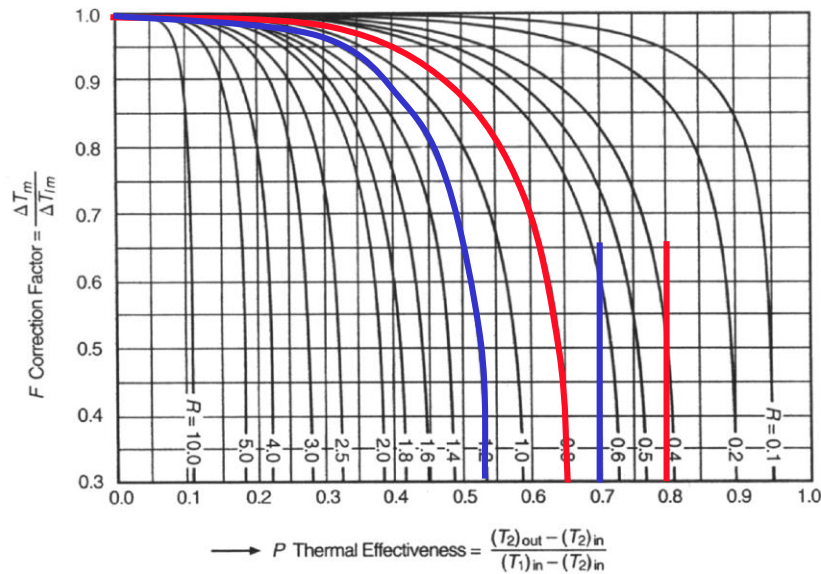
Cuando se produce temperatura de cruce la F suele tener un valor muy bajo. Para aumentar esta eficiencia manteniendo las condiciones de operación se usan configuraciones en serie



Temperatura de cruce



Cambio de los pasos ¡Cuidado con el encuentro de las dos temperaturas en la misma posición! Esfuerzo térmico



1 paso por la carcasa

El factor de corrección F es una medida de la eficiencia (comparado con un contracorriente) de un intercambiador. En el caso de intercambiadores tubo carcasa una F mínima aceptable es $F=0.8$ (0.9)

El aumento del nº de carcassas mejora la eficiencia relativa del intercambiador (mejor F) pero aumenta el coste al tener que adquirir varias unidades

4 pasos por la carcasa

